

2

Refraction of Light

In this chapter, we will learn about

- refraction of light at the boundary between two media
- some examples of refraction of light
- total internal reflection
- some examples of total internal reflection

2.1 Refraction

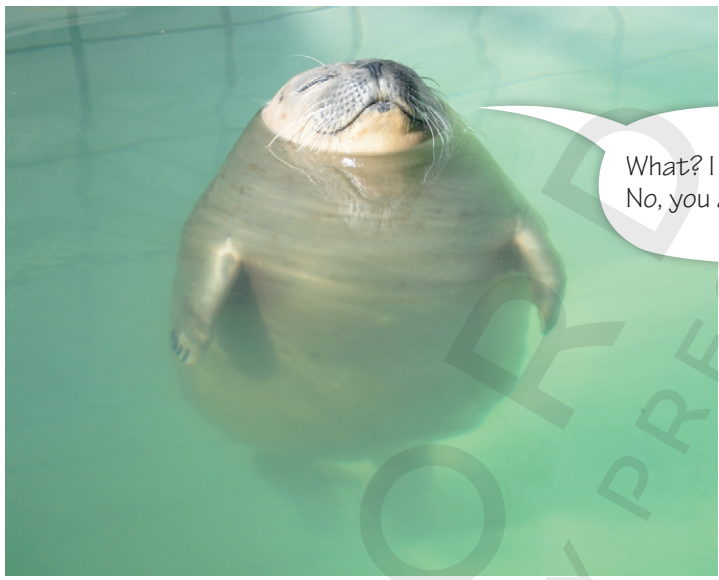


Fig 2.1a *A seal in water.*

In fact, the seal is not as fat as it looks. It appears like this because light bends when travelling from water to air. Let us learn more about the bending of light in this chapter.

1 Bending of light

In Figure 2.1b, a light ray bends when travelling from air into water. This bending of light is called **refraction**. Air and water are each called a **medium**.

Some light is reflected when light travels from one medium to another. We shall ignore this reflected ray and focus on the refracted ray in this chapter.

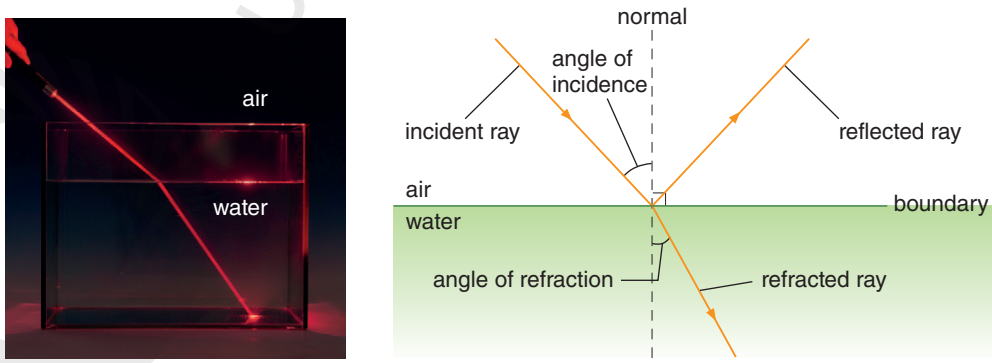


Fig 2.1b *Bending of light at the boundary between air and water.*

The following terms describing refraction are new to you (Table 2.1a).

Caution
The angle of refraction is **not** the angle between the refracted ray and the boundary surface.

Refracted ray	The ray that crosses the boundary after refraction
Angle of refraction	The angle between the refracted ray and the normal

Table 2.1a *Terms describing refraction.*

refraction 折射　medium (pl. media) 介質　refracted ray 折射線　angle of refraction 折射角

2 Speed of light and refraction

Why does a ray of light bend when it travels from one medium to another? This can be explained by the change in speed when light crosses a boundary. Try the following Hands-on activity.

Video

→ This video demonstrates the Hands-on activity.

Analogy for the refraction of light

Hands-on activity

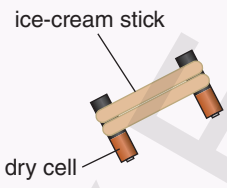
→ Worksheet is available on OUP web.

What you need



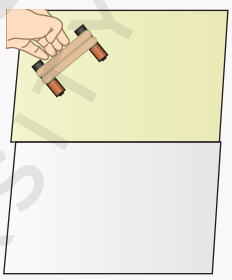
What to do

1



Fix the ice-cream sticks on the dry cells with adhesive tape to build a sledge. No tape should be attached to the base of the sledge.

2



Fix the two plastic sheets side by side on the table. Give a sharp push to the sledge so that it slides from the frosted sheet into the clear sheet at an angle. Repeat by pushing the sledge in the opposite direction.

What you find

How does the path of the sledge change when moving from one plastic sheet to another?

In the above activity, when the sledge is crossing the boundary, the batteries on the two sides move with different speeds. As a result, its path bends (Fig 2.1c and 2.1d).

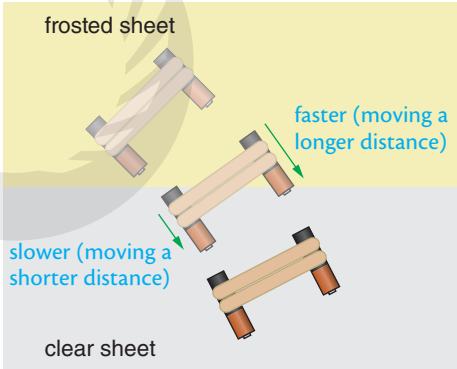


Fig 2.1c *The sledge bends after crossing the boundary.*

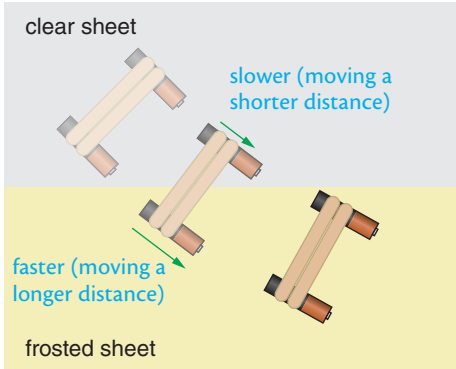


Fig 2.1d *The sledge bends to the other side after crossing the boundary.*

Light behaves similarly (Fig 2.1e).

For a light ray travelling from one medium to another at an angle,

- **it bends towards the normal if the speed of light decreases;**
- **it bends away from the normal if the speed of light increases.**

Light travels fastest in a vacuum (or air) and slower in other media.

Strictly speaking, light travels a bit slower in air than in a vacuum. However, unless otherwise specified, we assume that light travels at the same speed in these two media for simplicity.

Light is reflected at a boundary. The angle of reflection is equal to the angle of incidence.

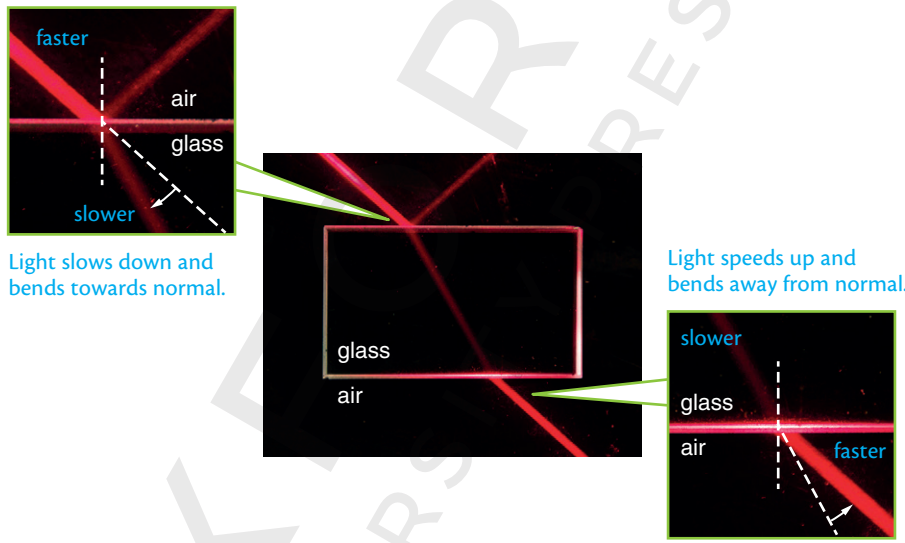


Fig 2.1e *Light ray passing through a glass block.*

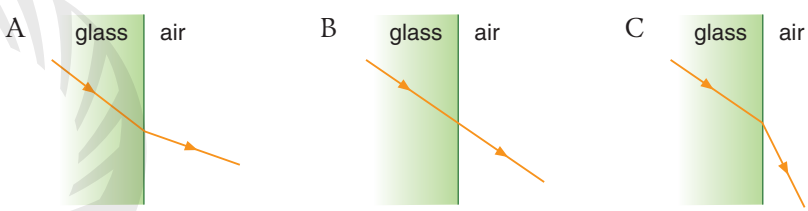
Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Refraction of light is caused by the change in speed of light when it travels from one medium to another. (T/F)

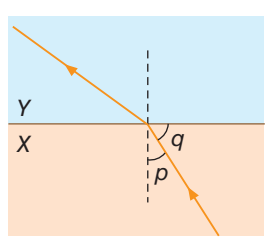
2 Which of the following diagrams shows a possible path for the light ray? [Hint: Draw the normal.]



3 A ray of light travels from media *X* to *Y* as shown.

(a) Which angle, *p* or *q*, is the angle of incidence?

(b) In which medium does light travel faster?



Know more

Refraction in atmosphere

Refraction of light also occurs in the atmosphere.

▸ p.77

3 Daily examples of refraction

Many daily phenomena can be explained after knowing how light bends. For example, a swimming pool filled with water appears shallower than it really is (Fig 2.1f).

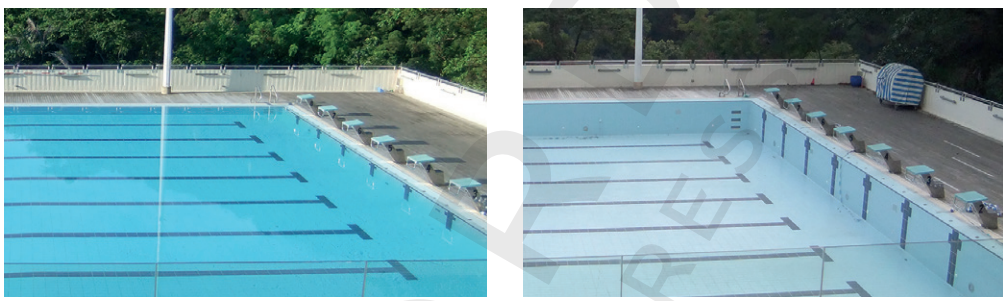


Fig 2.1f *Swimming pool (i) with and (ii) without water.*

→ This video shows how a bucket appears to be shallower when filled with water.

Simulation Video

→ This simulation shows (a) the apparent depth of a fish in water as viewed from air, and (b) the apparent height of a bird as viewed from underwater. The simulation is available on OUP web in PC version (.exe) only.

The seal on p.36 looks fat for the same reason. ▶

Consider a point *O* at the bottom of the swimming pool (Fig 2.1g). Due to refraction, the rays from *O* bend away from the normal at the water-air interface and appear to come from point *I*. We see a virtual image of *O* there. As a result, the pool appears shallower. The depth as seen by the observer is called the *apparent depth* and is smaller than the real depth in this case.

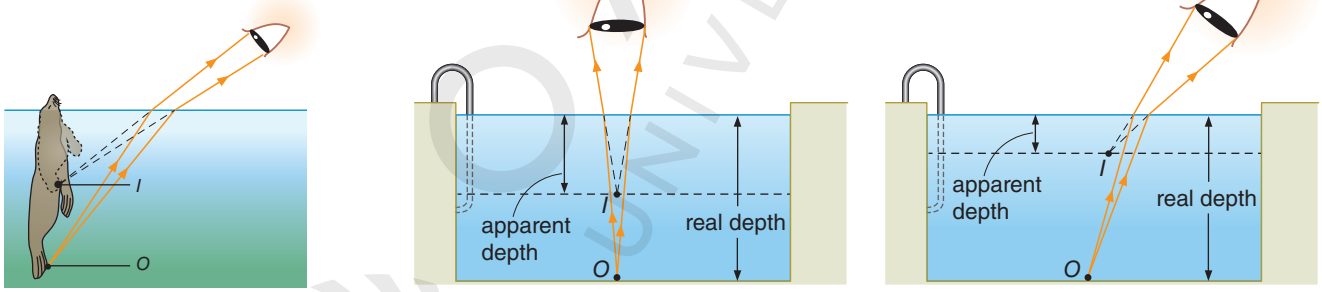


Fig 2.1g *Ray diagrams showing why a swimming pool filled with water appears shallower.*

On the other hand, if you are underwater and look at an object above water, the object will appear higher than its actual position (Fig 2.1h).

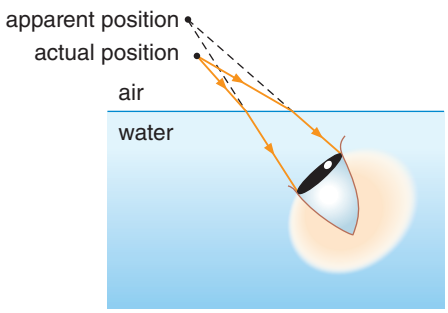
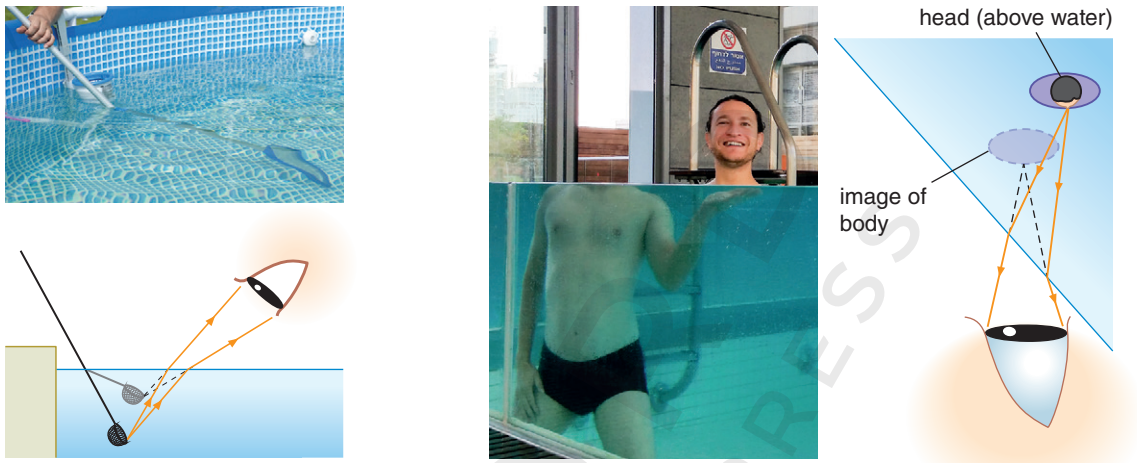


Fig 2.1h *Looking at an object from underwater.*

apparent depth 視深

Refraction also explains why some objects look broken when they are partly immersed in water (Fig 2.1i).



(i) A net pole partly immersed in water. *(ii) A man in a swimming pool with a glass wall.*

Fig 2.1i *Objects appear broken due to refraction.*

Recall that the speed of light in air is faster than that in water as well as other media except vacuum.

▶ Same effect can be found in media other than water (Fig 2.1j).



Fig 2.1j *The right part of the photo appears raised when it is put under a thick glass block.*

STSE

Solar bottle bulb

In *slums*, houses usually have few windows, so it is dark indoors even during daytime. To solve this problem, a water-filled soft drink bottle can be sealed into the roof (Fig a). By refraction, more sunlight enters the house (Fig b). These 'solar bottle bulbs' provide an economical and environmentally friendly way to light up the houses in daytime.



Fig a *A solar bottle bulb.*

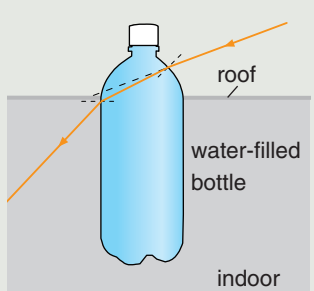


Fig b *Sunlight originally blocked can now enter the house.*

✎ A global project, called the Liter of Light, promotes the usage of solar bottle bulbs in places with limited access to electricity. The project was first carried out in the Philippines and served over 28 000 homes in Metro Manila. The information about the project can be found in the following website:

✎ A typical solar bottle bulb is as bright as an incandescent bulb of about 55 W.

slum 貧民窟

DSE goal

Examine the laws of refraction.

→ This simulation is a 'virtual experiment' of Snell's law.

→ This video demonstrates Expt 2a.

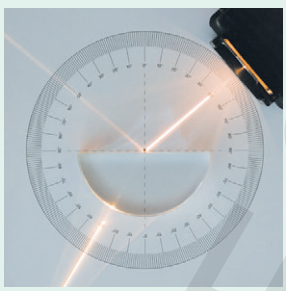
Simulation
Video

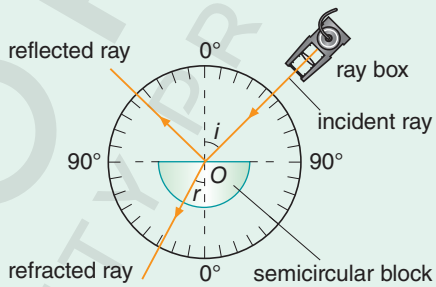
4 The laws of refraction

Let us look into refraction more closely. What is the relationship between the angle of incidence i and the angle of refraction r? In 1621, a Dutch scientist *Willebrord Snell* found the answer. Experiment 2a verifies his finding.

Experiment 2a / Snell's law

1 Set up the apparatus as shown. Measure the angle of incidence i and angle of refraction r.





2 Repeat step 1 with other values of i.

3 Plot a graph of sin i against sin r.

Alternatively, find the value of $\frac{\sin i}{\sin r}$ for each pair of i and r.

Discussion

1 What is the relationship between sin i and sin r?

2 State a source of error in the experiment.

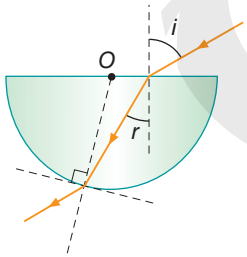
Do not touch the hot ray-box lamp.

Skill *Mathematics*

Circle, tangent and normal ▸ p.135

To guarantee that the light ray always hit the block's centre, Ts may consider to cover the straight edge of the semi-circular block with masking tape and leave a gap at the centre.

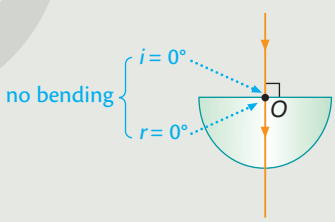
If the light ray does not hit O, it will be refracted at the curved edge, and the measured value of r will be different from the true value.

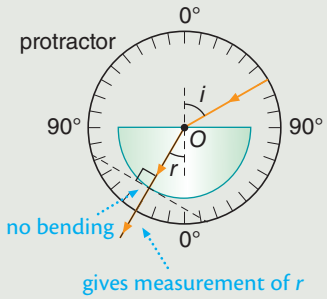


Know more

Semicircular block

In the experiment, when $i = 0°$, $r = 0°$. This suggests that a ray does not bend when it hits a boundary perpendicularly. Since a ray **through the centre O of the block always hits the curved edge perpendicularly**, it does not bend when crossing the curved edge. This allows us to read the value of r from the protractor.





Willebrord Snell 斯涅耳

Skill *Mathematics*

Straight line through origin ▸ p.136

The graph of sin i against sin r is a straight line passing through the origin (Fig 2.1k). This shows that

$$\frac{\sin i}{\sin r} = \text{constant}$$

This relationship is known as **Snell's law**. The constant depends on the two media which the light travels across.

A light ray does not bend when it passes from one medium to another perpendicularly. Therefore, when $i = 0°$, $r = 0°$. ▶

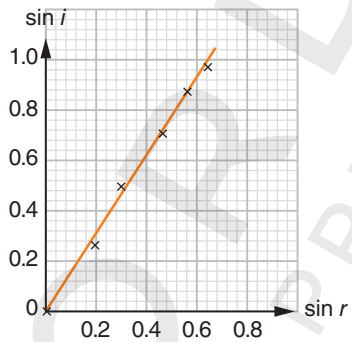


Fig 2.1k *A graph of sin i against sin r.*

Snell's law forms a part of the **laws of refraction**, which state that

When a ray of light passes from one medium to another,

- **the incident ray, the refracted ray and the normal all lie in the same plane.**
- $\dfrac{\sin i}{\sin r} = \text{constant}$

Know more

Willebrord Snell (1580–1626)

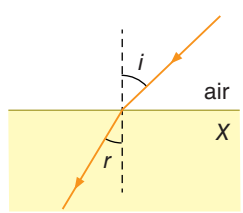


Snell was a Dutch astronomer and mathematician. In 1621, after examining a lot of data and much trial and error, he discovered that the ratio of sin i and sin r was constant, so this law was named after him. However, Snell did not develop a theory to arrive at this result.

OUP eResources

Example 1 | Applying Snell's law

A light ray enters medium X from air as shown. When $i = 15°$, $r = 10°$.



(a) Find $\dfrac{\sin i}{\sin r}$.

(b) Find r when $i = 60°$.

Solution

(a) $\dfrac{\sin i}{\sin r} = \dfrac{\sin 15^\circ}{\sin 10^\circ} = 1.49$

(b) By Snell's law,

$$\frac{\sin 60^\circ}{\sin r} = \frac{\sin 15^\circ}{\sin 10^\circ}$$

$$r = 35.5°$$

Snell's law 斯涅耳定律　　laws of refraction 折射定律

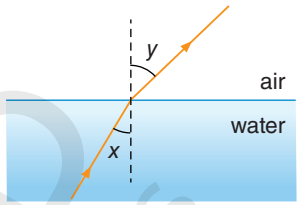
Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 A light ray travels from water to air as shown. Complete the table below.

	x	y
	30°	41.7°
(a)	20°	
(b)		30°



DSE goal

Realize $n = \frac{\sin i}{\sin r}$ as the refractive index.

DSE exam

13(1A)Q20, 14(1B)Q5(a), 18(1A)Q17

5 Refractive index

Consider a light ray crossing the boundary between **air** and medium X (Fig 2.1l). By Snell's law, the ratio of sin θ_a to sin θ_X is constant. This ratio is known as the **refractive index** n of the medium.

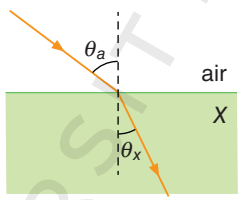


Fig 2.1l A light ray travelling from air to medium X.

In measuring the refractive index, it does not matter whether the light travels from air to medium X or vice versa. ▶

refractive index of X (no unit) $n_X = \frac{\sin \theta_a}{\sin \theta_X}$

angle between ray and normal in air (°)

angle between ray and normal in X (°)

Strictly speaking, n is obtained in a vacuum instead of air, but it makes very little difference. ▶

Simulation

→ This simulation shows how a light ray is refracted when it passes from one medium to another.

✎ Ts may further explain that the refractive indices are measured at a certain wavelength (589 nm) and air is measured at STP.

The refractive index has no unit. Different media have different refractive indices (Table 2.1b). Unless otherwise specified, we will not distinguish air from a vacuum and take the refractive index of air as 1.





Medium	Refractive index
Vacuum	1 (exact)
Air	1.0003 ≈ 1
Water	1.33
Vegetable oil	1.47
Perspex (Acrylic)	1.50
Glass	1.50–1.70
Diamond	2.42



Table 2.1b Refractive indices of some media.

refractive index (pl. indices) 折射率

Figure 2.1m shows rays of light travelling from air to different media with the same θ_a (60°). The greater the refractive index, the more the light bends.

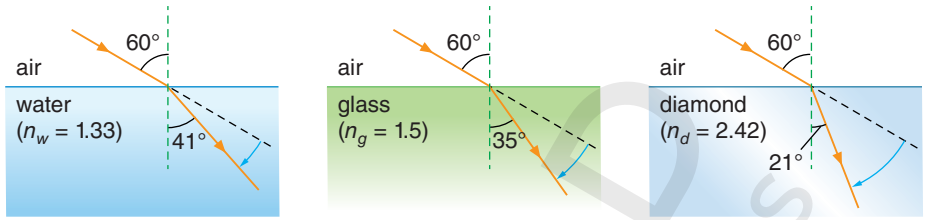


Fig 2.1m *Refraction in different media.*

Caution
The optical density of a medium is **different** from the density (or mass density).

For two media, the one with a greater refractive index is said to be **optically denser**, while that with a lower refractive index is **optically less dense**. For example, glass is optically denser than water.

Example 2 | Refractive index

Ts may emphasize the reasons of plotting graph:
(i) Checking data
(ii) Averaging error
(iii) Predicting values

Connie directs a ray of light from glass to air (Fig a). She then plots a graph of sin r against sin i (Fig b).

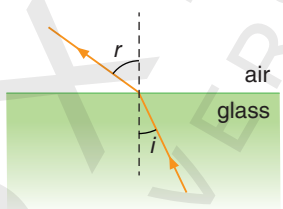


Fig a

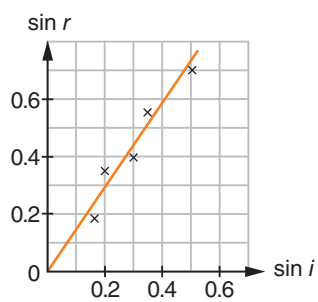


Fig b

(a) Find the refractive index n_g of the glass.

(b) Connie directs a ray of light from medium X to air. The refractive index of X is smaller than that of glass. Sketch the graph of sin r against sin i in Figure b.

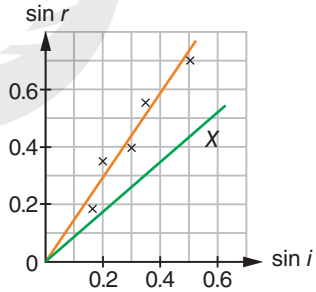
Tip
The slope of the graph instead of one data point is used to calculate n. This averages out the error in each measurement.
Daily-life media have refractive indices greater than 1. You should check your answer if you have $n < 1$.

Solution

(a) $n_g = \dfrac{\sin \theta_a}{\sin \theta_g} = \dfrac{\sin r}{\sin i} = \text{slope} = \dfrac{0.6 - 0}{0.4 - 0} = 1.5$

Tip
$\because n$ = slope
$n \downarrow \Rightarrow$ slope $\downarrow$

(b)



Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

(For Q1–2.) A ray of light travels from air to medium X (Fig a).

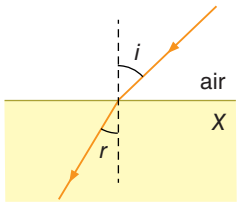


Fig a

1 Complete the table.

	i	r	n_X
(a)	20°	10°	
(b)	30°		1.6
(c)		30°	1.3

2 Medium X is replaced by media P and Q in turn and the corresponding graphs of sin r against sin i are shown (Fig b).

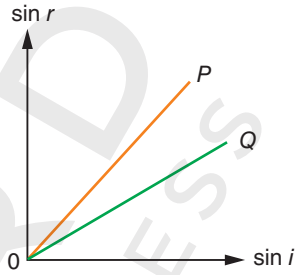


Fig b

Which medium has a greater refractive index?

3 Which medium, air, water or diamond, has the smallest refractive index?

Learn by practice 1

1 A ray of light travels from air to water. The angle of incidence is 40°. The refractive index of water is 1.33. Find the angle of refraction.

2 A ray of light travels from glass to air. The angle of incidence is 20° and the angle of refraction is 30.9°.

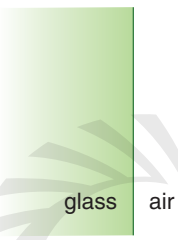


Fig a

(a) Draw the incident ray, normal and refracted ray in Figure a.

(b) Find the refractive index of the glass.

3 In an experiment, a student directs a light ray from air to glass and obtains the results as shown below.

θ_a	15°	30°	45°	60°	75°
θ_g	11°	18°	28°	34°	40°

(a) Plot a straight-line graph relating sin θ_a and sin θ_g.

(b) Estimate the refractive index of the glass from the graph in (a).

(c) The experiment is repeated by replacing the glass with another medium of a greater refractive index. Sketch the corresponding graph for this medium on the graph in (a).

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DSE goal

Sketch path of a ray refracted at boundary.

Solve problems involving refraction.

6 General form of Snell's law

In general, Snell's law can be expressed as

refractive index of medium 1 (no unit) — **refractive index** of medium 2 (no unit)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

angle between ray and normal in medium 1 (°) — **angle** between ray and normal in medium 2 (°)

If medium 1 is air, $n_1 = 1$, and the equation becomes

$1 \times \sin \theta_a = n_2 \sin \theta_2$

$$n_2 = \frac{\sin \theta_a}{\sin \theta_2}$$

which is identical to the equation on p.43.

Note that (Fig 2.1n)

(i) if $n_2 > n_1$, $\theta_2 < \theta_1$ (light bends towards the normal after refraction)

(ii) if $n_2 < n_1$, $\theta_2 > \theta_1$ (light bends away from the normal)

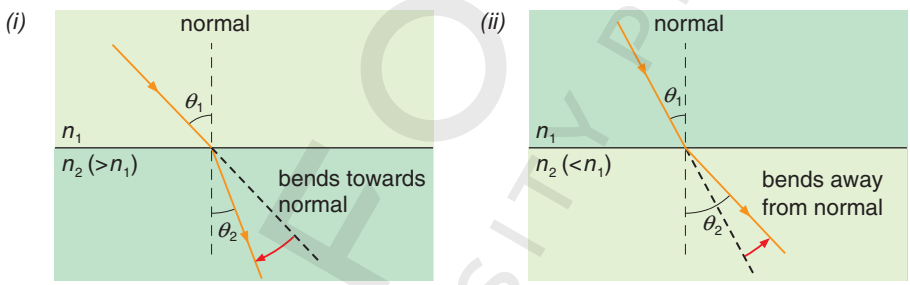


Fig 2.1n *How a light ray bends when (i) $n_2 > n_1$ and (ii) $n_2 < n_1$.*

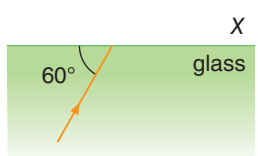
Example 3 General form of Snell's law

A ray of light travels from glass to medium X. The refractive index of glass is 1.50. Find the angle of refraction if X is

(a) air;

(b) water (refractive index = 1.33);

(c) oil (refractive index = 1.47).

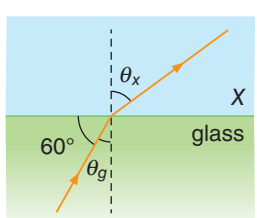


Solution

Caution

Do **not** take $\theta_g = 60°$.

$\theta_g = 90° - 60° = 30°$



Apply $n_g \sin \theta_g = n_X \sin \theta_X$.

(a) $1.50 \times \sin 30° = 1 \times \sin \theta_a$

$\theta_a = 48.6°$

Caution

$n_g \neq \frac{\sin \theta_w}{\sin \theta_g}$

(b) $1.50 \times \sin 30° = 1.33 \times \sin \theta_w$

$\theta_w = 34.3°$

(c) $1.50 \times \sin 30° = 1.47 \times \sin \theta_o$

$\theta_o = 30.7°$

Know more

Water beads

Water beads can be used for planting. When a bead fully absorbs water, its refractive index becomes very close to that of water. This makes it almost invisible in water.



Video

→ This video shows how the balls emerge when the water is poured out.

In part (c) of Example 3, the light ray bends by 0.7° only when travelling from glass to oil. Indeed, if the difference in reflective index is very small, light barely bends. In this case, the refraction and reflection of light at the boundary can hardly be noticed. This explains why glass objects are almost invisible in oil (Fig 2.1o).



Fig 2.1o Glass bowl in oil.

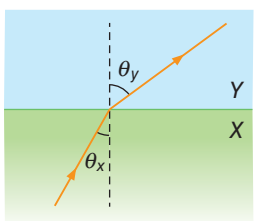
Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 A ray of light travels from media X to Y (Fig a).

Complete the table below.



	θ_X	θ_Y	n_X	n_Y
(a)	10°	14°	1.7	
(b)	20°		1.3	1.8
(c)		50°	1.3	1.5

Learn by practice 2

1 A ray of light travels through media P, Q and R (Fig a).

(a) Which is greater, n_P or n_Q?

(b) Which is greater, n_Q or n_R?

(c) Arrange n_P, n_Q and n_R in ascending order.

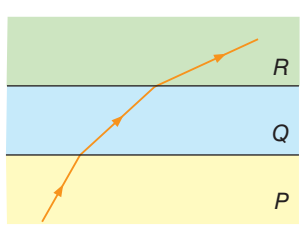


Fig a

2 A ray of light travels from water into a diamond (Fig b). The refractive indices of water and diamond are 1.33 and 2.42 respectively. Find the angle of refraction and sketch the refracted ray in the diamond.

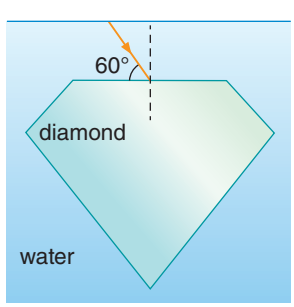


Fig b

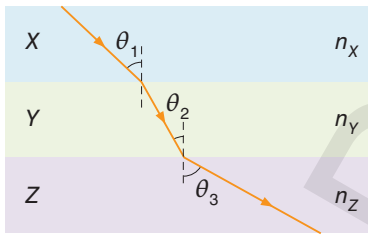
☆ HKDSE 2016 Paper 1A Q17

→ This video demonstrates how this question is solved.

Video

DSE practice 1 Light passing through multiple layers of materials

Three parallel-sided glass plates X, Y and Z of refractive indices n_X, n_Y and n_Z are joined together. A ray of light passes through the plates as shown.



Which of the following descriptions is/are correct?

(1) $n_Z > n_X > n_Y$

(2) θ_3 is independent of n_Y.

(3) If Y and Z swap position, the ray emerging into Y will bend away from the normal.

A (2) only

B (3) only

C (1) and (3) only

D (1), (2) and (3)

Solution

Consider the X–Y interface. The ray bends towards the normal, so $n_X < n_Y$.

∴ (1) is incorrect.

Consider the X–Y interface. By Snell's law,

$n_X \sin \theta_1 = n_Y \sin \theta_2$ ..... (1)

Consider the Y–Z interface. By Snell's law,

$n_Y \sin \theta_2 = n_Z \sin \theta_3$ ..... (2)

Substitute (1) into (2),

$n_X \sin \theta_1 = n_Z \sin \theta_3$

This shows that θ_3 is independent of n_Y.

As long as the interfaces are parallel.

∴ (2) is correct.

Common mistake
Students may wrongly think that θ_3 depends on n_Y. In fact, θ_3 is independent of n_Y if the X–Y interface and Y–Z interface are parallel.

The figure shows that $\theta_3 > \theta_2$ which implies $n_Z < n_Y$. If Y and Z swap positions, the ray will travel from an optically less dense medium (Z) into an optically denser medium (Y). Therefore the ray emerging into Y will bend towards the normal.

∴ (3) is incorrect.

∴ The answer is A.

▶ Revision 2 Q12 (p.70)

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DSE goal

Define refractive index in terms of wave speeds.

DSE exam

14(1A)Q15, 20(1A)Q13

7 Speed of light and refractive index

Light travels at speed c (3.00×10^8 m s^{-1}) in air and at lower speed v in any another medium. The refractive index of a medium is defined as

refractive index of medium (no unit) $n = \dfrac{c}{v}$ **speed** of light in air (m s^{-1}) / **speed** of light in medium (m s^{-1})

The **lower the speed of light** in a medium, the **greater the refractive index** of the medium (optically denser). Figure 2.1p shows the speed of light in some media and their corresponding refractive indices.

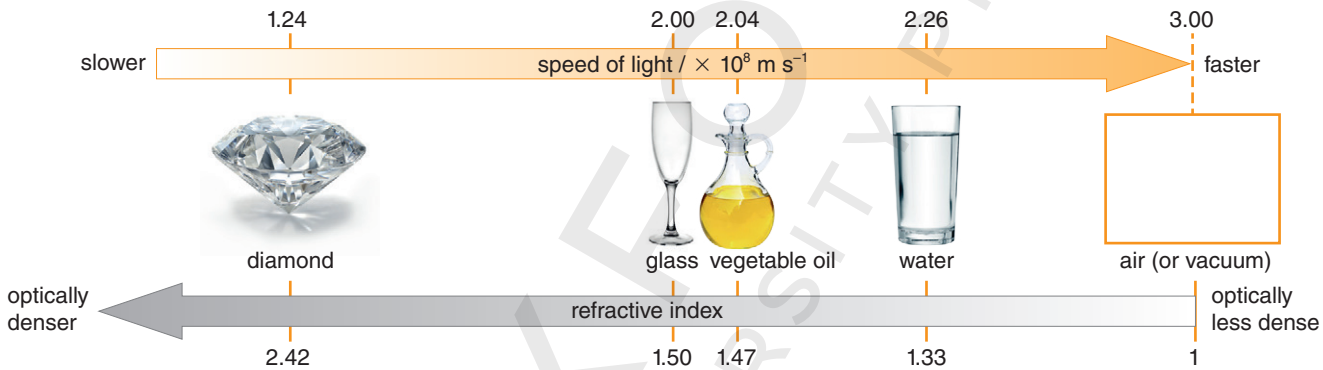


Fig 2.1p *Speeds of light in media with different refractive indices.*

Example 4 Finding the speed of light in different media

(a) The speed of light in air is 3.00×10^8 m s^{-1} and the refractive index of water is 1.33. Find the speed of light in water.

(b) When a light ray travels from medium X to medium Y, its speed changes from v_X to v_Y. The angle of incidence and the angle of refraction are θ_X and θ_Y respectively. Show that $\dfrac{v_Y}{v_X} = \dfrac{\sin \theta_Y}{\sin \theta_X}$.

Solution

$n = \frac{c}{v}$ ▶
$v = \frac{c}{n}$

(a) Speed of light in water $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8}{1.33} = 2.26 \times 10^8$ m s^{-1}

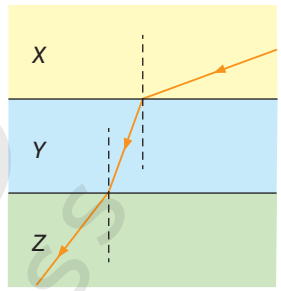
(b) $n_X \sin \theta_X = n_Y \sin \theta_Y$

$$\frac{c}{v_X} \times \sin \theta_X = \frac{c}{v_Y} \times \sin \theta_Y$$

$$\frac{v_Y}{v_X} = \frac{\sin \theta_Y}{\sin \theta_X}$$

Example 5 Speed of light in different media

Light travels through media X, Y and Z as shown. The boundaries between the media are parallel.



(a) Which medium has the smallest refractive index? Explain.

(b) In which medium does light travel the fastest? Explain.

Solution

(a) Since $n_X \sin \theta_X = n_Y \sin \theta_Y = n_Z \sin \theta_Z$,

$\theta_X > \theta_Z > \theta_Y \Rightarrow n_X < n_Z < n_Y$

$\therefore$ X has the smallest refractive index.

(b) Since $n = \frac{c}{v}$,

$n_X < n_Z < n_Y \Rightarrow v_X > v_Z > v_Y$

$\therefore$ Light travels the fastest in X.

From (a), $\theta_X > \theta_Z > \theta_Y \Rightarrow v_X > v_Z > v_Y$

Google Forms Microsoft Forms

Quick check 5

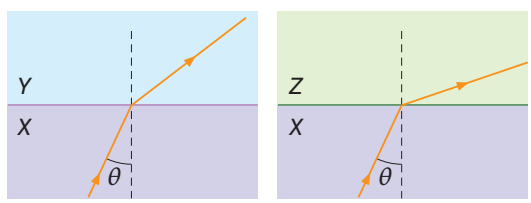
→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Light travels faster in optically denser medium. (T/F)

2 Light travels at 3.00×10^8 m s^{-1} in air. It travels at v in a medium. Complete the table.

	n	v / m s^{-1}
(a)		2.00×10^8
(b)	2.40	

3 In which of the medium below does light travel the fastest?



Learn by practice 3

1 The refractive index of sapphire (Fig a) is 1.77. Find the speed of light in it. The speed of light in air is 3.00×10^8 m s^{-1}.



Fig a

2 Light travels the slowest in medium P and the fastest in medium Q. Sketch in Figure b the path of the light ray.

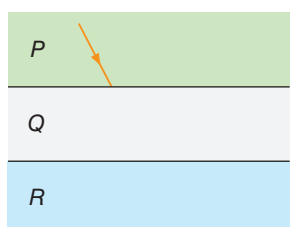


Fig b

DSE exam

13(1A)Q21, 14(1B)Q5(d), 16(1A)Q20

8 Dispersion

When a beam of white light passes through a prism, it is split into the colours of a rainbow (the *colour spectrum*) (Fig 2.1q). This effect is called **dispersion**.

The colour spectrum is also called the visible spectrum.

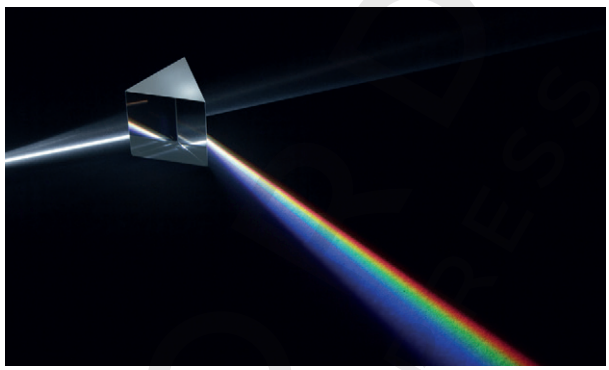


Fig 2.1q *White light is dispersed into different colours by a glass prism.*

Simulation

→ This simulation demonstrates the dispersion of light. Ss can control the wavelength and angle of incidence of the light ray entering the prism. The simulation is available on OUP web in PC version (.exe) only.

White light is composed of a mixture of a whole range of coloured lights. Different coloured lights travel with the same speed in air. However, when they enter another medium like glass or water, they slow down to a slightly different extent and are refracted at slightly different angles (Fig 2.1r). In other words, they have **different refractive indices** in the medium.

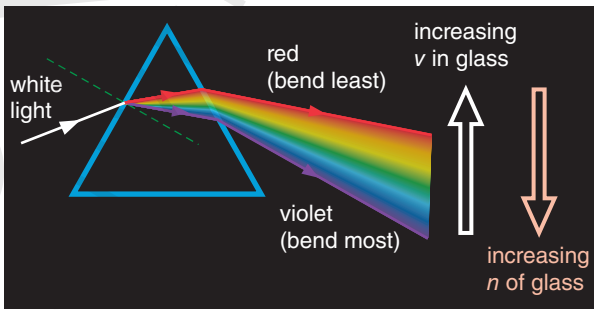


Fig 2.1r *Different colours of light are refracted at slightly different angles after passing through a glass prism.*

Know more

Newton and the colour spectrum

In 1666, Newton directed sunlight from a hole to pass through a glass prism and obtained a colour spectrum. He then concluded that white light is composed of a mixture of a whole range of colours. This finding of Newton has led to breakthroughs in the study of optics and the nature of colour.



Rainbow

The appearance of a rainbow is caused by the dispersion of sunlight in water droplets.

▸ p.78

colour spectrum 色譜　　dispersion 色散

We can see that **violet light bends the most** and **red light bends the least** upon refraction. Therefore, **red light** has a **higher speed** and a **smaller refractive index** than violet light in glass.

Different coloured lights travel at different speeds in a prism and hence have different refractive indices.

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Red light travels faster than violet light in glass. (T/F)

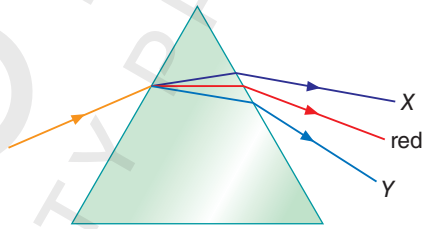
(b) Red light travels faster than violet light in vacuum. (T/F)

(c) Red light has a greater refractive index than violet light in glass. (T/F)

(d) When white light travels from an optically denser medium to a less dense one, there is no dispersion. (T/F)

(e) When white light travels from air to glass perpendicularly, there is no dispersion. (T/F)

2 A ray of white light passes through a prism. Which path, *X* or *Y*, is a possible path for violet light?



Exercise 2.1

Take the speed of light in air as 3.00×10^8 m s^{-1}.

1 A ray of light is directed from air into a semicircular glass block. As θ varies, ϕ is measured (Fig a).

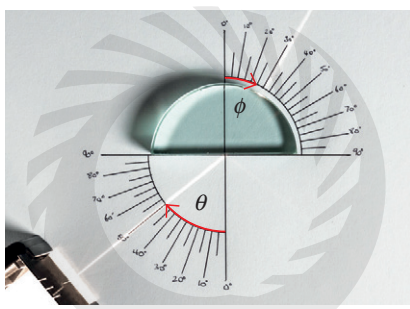


Fig a

Which of the following graphs has a slope denoting the refractive index of the glass?

A θ against ϕ

B ϕ against θ

C $\sin\theta$ against $\sin\phi$

D $\sin\phi$ against $\sin\theta$

★**2** The dispersion of white light in a glass block is shown below (Fig b). *X* and *Y* are the two ends of the colour spectrum projected on paper.

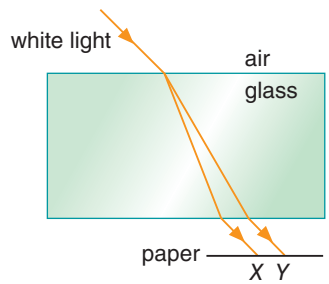


Fig b

Which of the following statements is/are correct?

(1) *X* is the red end of the spectrum.

(2) *X* travels faster than *Y* in glass.

(3) The beam emerging from the block is parallel.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★**3** Light travels at 2.0×10^8 m s^{-1} in medium X. A ray of light travels from X to air as shown (Fig c). What is the angle of refraction?

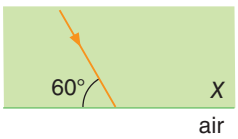


Fig c

A 19.5°

B 25.7°

C 35.3°

D 48.6°

★**4** A ray of red light passes through a 60°–60°–60° prism as shown (Fig d). What is the speed of the red light in the prism?

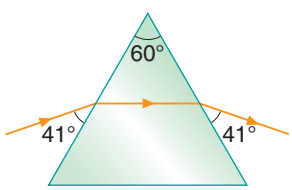


Fig d

A 1.99×10^8 m s^{-1}

B 2.29×10^8 m s^{-1}

C 2.61×10^8 m s^{-1}

D 3.96×10^8 m s^{-1}

★**5** The figures below show a ray of light travels from air to glass (Fig e) and from glass to air (Fig f). Given: n of glass = 1.50.

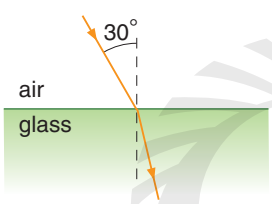


Fig e

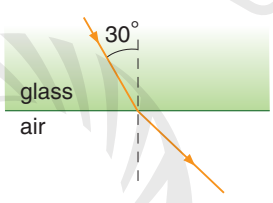


Fig f

In each figure, sketch the new refracted ray if the glass is replaced by

(a) water of n = 1.33;

(b) crystal of n = 1.70.

★**6** Two rays of light p and q are incident on a water-glass interface (Fig g). Complete the path of q in the glass.

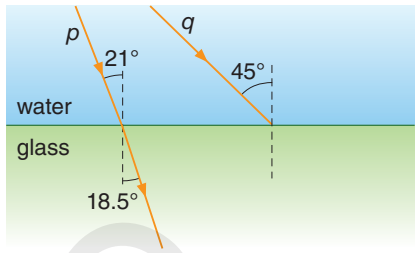


Fig g

★**7** A light ray travels from water to glass. The speeds of light in water and in glass are 0.752c and 0.645c respectively, where c is the speed of light in air. If the angle of refraction is 27°, what is the angle of incidence?

★**8** Two parallel rays A and B enter a circular glass block from air (Fig h). Complete the paths of the rays. Find all the angles the rays make with the normal at the air-glass interfaces. Note that the normal of a circle always passes through the centre of the circle. Given: n of glass = 1.50.

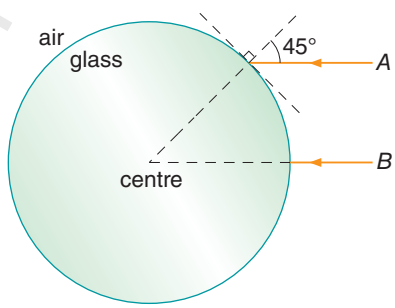


Fig h

★**9** A ray of light is directed from air into a semicircular transparent block at point P (Fig i). A graph of sin i against sin r is shown below (Fig j).

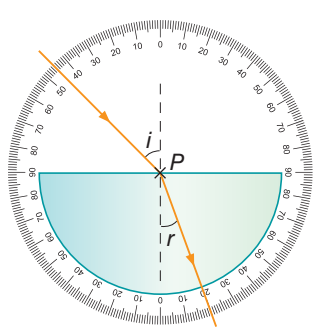


Fig i

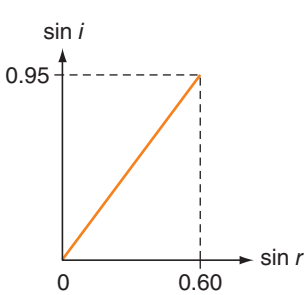


Fig j

(a) Find the refractive index of the block.

(b) Can we measure the true value of r if the light ray is not directed at point P?

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2.2 Total internal reflection

Try this: prepare two transparent plastic cups *A* and *B*. Draw a magician on *A* and a cage on *B* (Fig 2.2a). Insert *A* into *B*. Lower the cups into water until the water just covers the cage. The magician disappears!

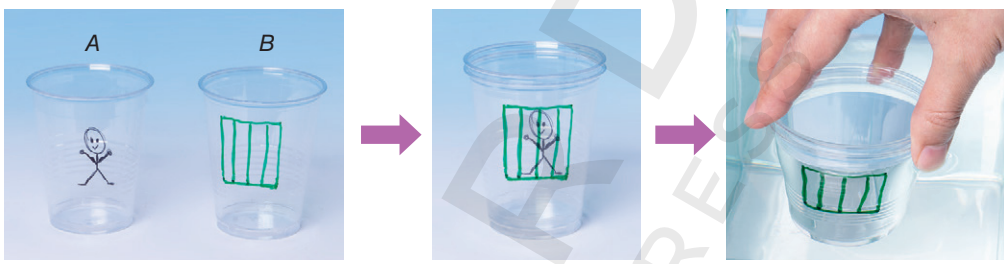


Fig 2.2a *Magic using plastic cups and water.*

Video

→ This video shows how this magic works.

See p.78 for details.

▶ This magic makes use of a phenomenon called **total internal reflection**.

DSE goal

Examine conditions for total internal reflection.

Solve problems involving total internal reflection.

DSE exam

14(1B)Q5(b)-(c), 16(1A)Q7, 17(1B)Q7(a), 21(1A)Q13, 23(1A)Q16, 23(1B)Q6

Simulation Video

→ This simulation is a 'virtual experiment' on total internal reflection. Ss can select from several media for investigation: ice, Perspex, glass, crystal and diamond. The simulation is available on OUP web in PC version (.exe) only.

Do not touch the hot ray-box lamp.

→ This video demonstrates Expt 2b.

→ This is a Video Quiz about this experiment.

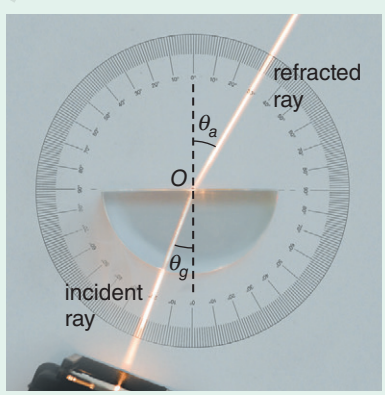
1 Criteria for total internal reflection

Experiment 2b / Total internal reflection

1 Put a semicircular glass block on a full circle protractor. The straight edge of the block must lie on the 90°-90° line and the centre *O* of the block at the centre of the protractor.

2 Direct a ray of light at *O* from the curved edge of the block.

3 Slowly increase θ_g and see how the refracted ray changes.



Discussion

1 How does θ_a change as θ_g increases?

2 What is the value of θ_g when θ_a reaches 90°?

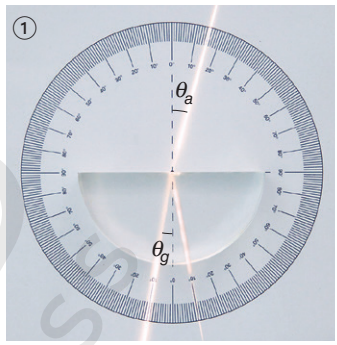
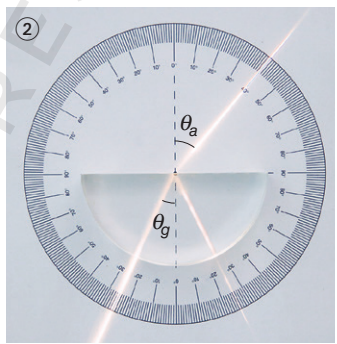
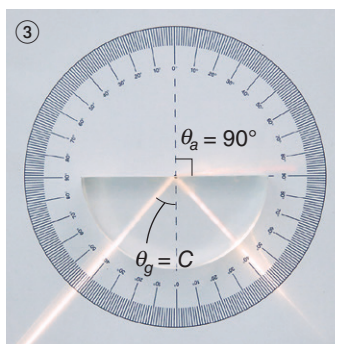
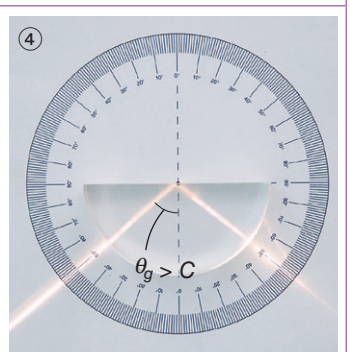
3 What happens to the refracted ray when θ_g is further increased?

total internal reflection 全內反射

Video

→ This video shows an experiment investigating total internal reflection in water using a laser pointer as a light source.

The results of Experiment 2b are as follows.

When the light ray hits the straight edge, part of it is reflected and part of it is refracted. The refracted ray bends away from the normal.	
As θ_g increases, θ_a increases.	
As θ_g reaches a certain value C, θ_a becomes 90°. The refracted ray just manages to leave the glass surface.	
If θ_g continues to increase, there is no more refracted ray. The straight edge acts like a mirror. **All the incident ray is reflected**. This phenomenon is called total internal reflection.	

As $\theta_a > \theta_g$, θ_a reaches 90° before θ_g does when θ_g keeps increasing.

The angle of reflection always equals the angle of incidence.

In ③, angle C is the **critical angle** of the glass-air interface. It can also be found by Snell's law (when $\theta_g = C$, $\theta_a = 90°$):

$$n \sin C = 1 \times \sin 90°$$

For a boundary between two media with refractive indices n_1 and n_2, where $n_1 > n_2$, we have

$$n_1 \sin C = n_2 \sin 90°$$

$$\sin C = \frac{n_2}{n_1}$$

critical angle of medium-air interface (°) — $\sin C = \frac{1}{n}$ — **refractive index** of medium (no unit)

critical angle 臨界角

For total internal reflection to occur, the angle of incidence should be larger than the critical angle C (Fig 2.2b). In addition, light should travel from glass (optically denser) to air (optically less dense). Both conditions must be satisfied.

Total internal reflection occurs when

1. **light travels from an optically denser medium to an optically less dense medium, and**
2. **the angle of incidence is larger than the critical angle C.**

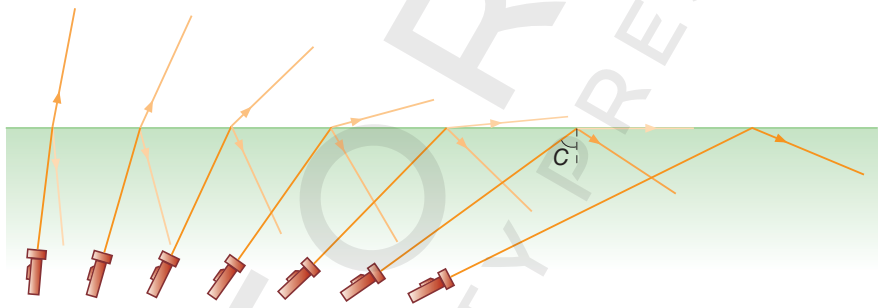


Fig 2.2b *Total internal reflection occurs only when the angle of incidence is larger than C.*

Know more

Total internal reflection tricks

You can perform some tricks using total internal reflection.



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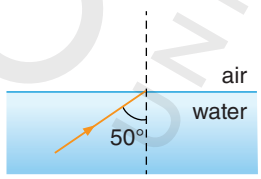
Example 6 | Total internal reflection

Water has a refractive index of 1.33.

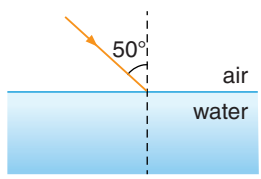
(a) Find the critical angle of a water-air interface.

(b) Will total internal reflection occur in the following cases?

(i)



(ii)



Solution

(a) $\sin C = \frac{1}{n} = \frac{1}{1.33}$

$C = 48.8°$

(b) (i) Yes

(ii) No

Tip

$i > C$

Tip

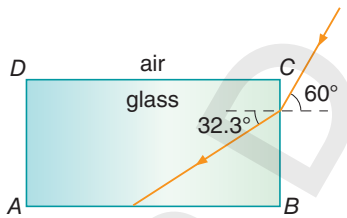
Total internal reflection **never** occurs when light travels from an optically less dense medium (air) to an optically denser medium (water).

Think about...

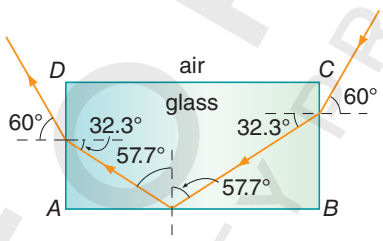
Will the answers to (b) change if glass of refractive index 1.5 is used instead of water?

Example 7 | Light ray through a block (I)

A ray of light hits a glass block as shown. Sketch the path of the light ray.



Solution



Tip

$n = \frac{\sin 60°}{\sin 32.3°} = 1.62$

$\sin C = \frac{1}{n} = \frac{1}{1.62}$

$C = 38.1°$

At side AB,

$i = 90° - 32.3°$

$= 57.7°$

$> 38.1°$

$\therefore$ Total internal reflection occurs.

Quick check 7

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

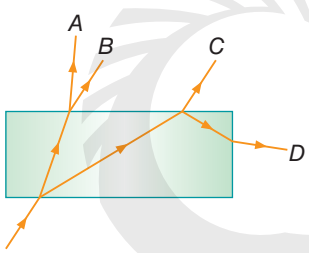
1 *True or false:*

(a) Total internal reflection will never occur if light travels from water ($n = 1.33$) to glass ($n = 1.5$). (T/F)

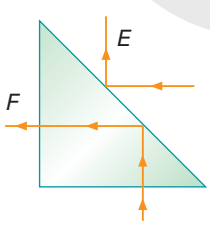
(b) The greater the refractive index of a medium, the larger the critical angle for the medium-air interface. (T/F)

2 In the following cases, light travels from air to glass blocks. Which light rays are possible?

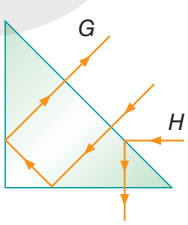
(a)



(b)



(c)

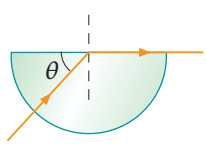


3 Consider a light ray travelling from the following media to air. Complete the table.

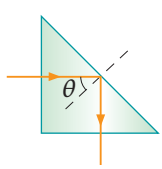
	n	C
Diamond		24.4°
Oil	1.47	

4 In which of the following figures does the angle θ indicate the critical angle for the interface?

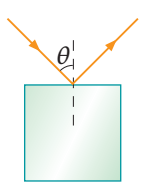
A



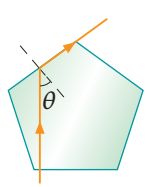
B



C



D



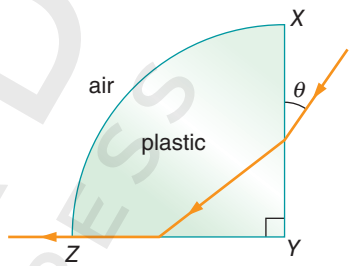
☆ HKDSE 2017 Paper 1B Q7

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Light ray through a block (II)

The cross-section of plastic block *XYZ* is a quarter circle with its centre at *Y*. An orange light ray hits the block from air as shown. The refractive index of the block for orange light is 1.3.



(a) Find the critical angle of the plastic-air interface for orange light. (2 marks)

(b) Find θ. (2 marks)

(c) A violet light ray hits the block at the same point at the same angle as the orange light ray. Violet light travels slower than orange light in the block.

(i) Compare the critical angle for orange and violet lights at the plastic-air interface. (1 mark)

(ii) Sketch the path of the violet ray. (2 marks)

Solution

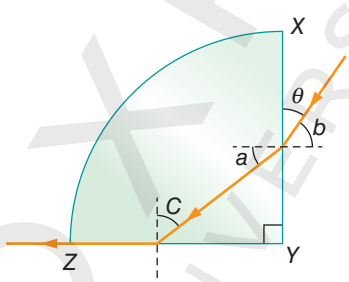
(a) $\sin C = \frac{1}{n} = \frac{1}{1.3}$ 1M

$C = 50.3°$ 1A

(b) $a = 90° - C$

$= 90° - 50.3°$

$= 39.7°$



At interface *XY*,

by $n_1 \sin \theta_1 = n_2 \sin \theta_2$, 1M

$1.3 \times \sin 39.7° = 1 \times \sin b$

$b = 56.2°$

$\theta = 90° - b = 33.8°$ 1A

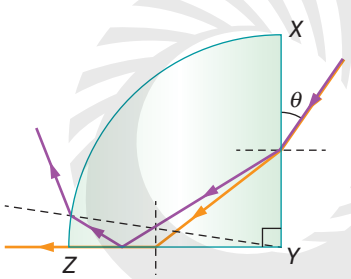
Common mistake
Students may wrongly take θ as the angle of incidence.

(c) (i) The critical angle for violet light is smaller. 1A

$\sin C = \frac{1}{n} = \frac{1}{\frac{c}{v}} = \frac{v}{c}$

$v_v < v_o \Rightarrow C_v < C_o$

(ii)



(Reflected ray correct at *YZ*, $i = r$) 1A

(Ray bends away from normal when leaving block) 1A

Common mistake
Students may not know that violet light undergoes total internal reflection at interface *YZ*. Some wrongly draw the ray straight through interface *XZ* without bending.

▶ Revision 2 Q29 (p.76)

DSE exam
15(1B)Q6

2 Examples of total internal reflection

a Mirage

On a sunny day, the ground a distance ahead may appear wet and reflective. In fact, the ground is dry. What we see is a *mirage* (Fig 2.2c).

Video

→ This video shows a mirage at the airport. An aeroplane is landing and an inverted image of it can be seen.



Fig 2.2c *Examples of mirages.*

Figure 2.2d illustrates what has happened. The ground, and hence the air near it, is heated up. Hot air at the bottom has a smaller refractive index than cool air at the top. As light passes through layers of air of increasing temperature, it is gradually refracted and becomes more horizontal. When the angle of incidence i into a certain layer is beyond the critical angle C, total internal reflection occurs.

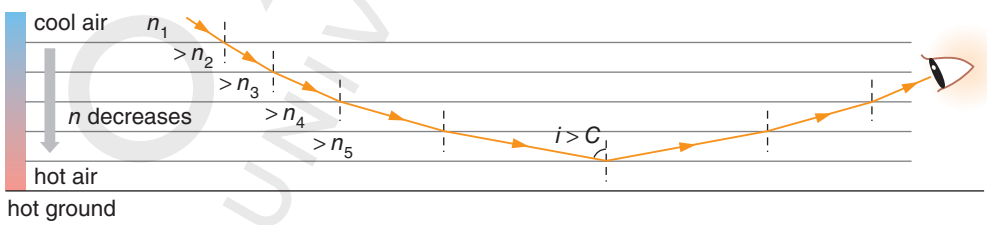


Fig 2.2d *The bending of light through layers of air of increasing temperature.*

Therefore, an observer sees the inverted images of distant objects (Fig 2.2e). The 'water' on the ground is actually the image of the sky!

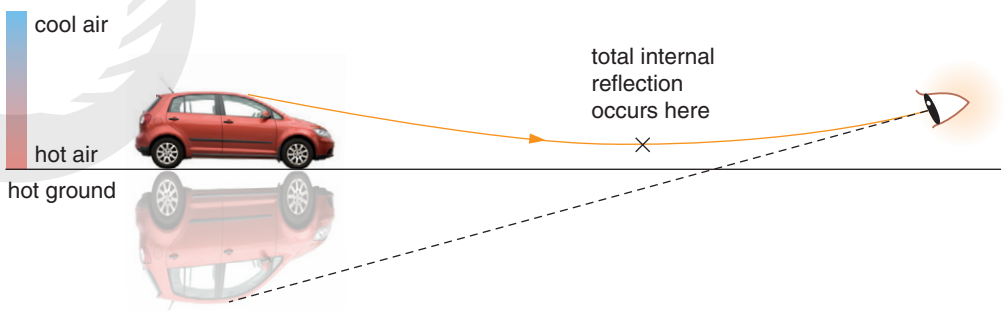


Fig 2.2e *An observer sees the inverted image of a distant object.*

mirage 海市蜃樓

Video → This video shows total internal reflection occurs in concentrated sugar solution. A gradation of density is produced in a water tank. A light beam is made to curve by refraction and then totally internally reflected in the solution.

Sometimes the temperature gradient is reversed, i.e. air is cooler at the bottom (e.g. polar seas). Light from a distant object is bent down when it passes through layers of air. As a result, an image of the object is seen at a position higher than the object (Fig 2.2f). This phenomenon is known as *superior mirage*. The set-up in Figure 2.2g simulates the situation.



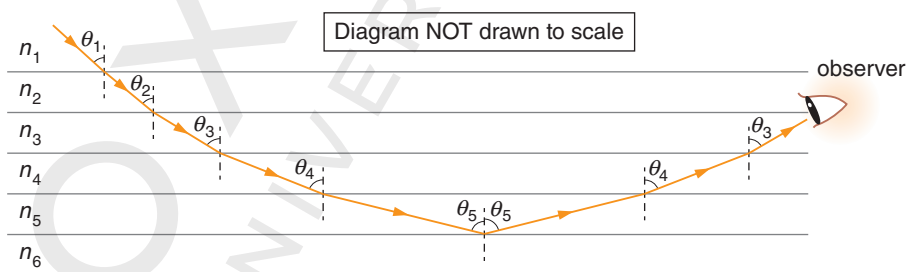
Fig 2.2f *A superior mirage occurs above a cold sea: an inverted image of a ship is formed above it.*



Fig 2.2g *Light beam entering the tank is gradually refracted and totally internally reflected.*

Example 8 Mirage

A ray of light from the sky travels through parallel layers of air of different refractive indices as shown.



(a) Find θ_1 if $\theta_5 = 90°$. It is given that $n_1 = 1.000\ 273$ and $n_5 = 1.000\ 233$.

(b) The observer sees the image of the sky on the ground in front of him. If he walks forwards, can he get closer to the image? Explain.

Solution

See DSE practice 1 on p.48. ▶

(a) $n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = n_4 \sin\theta_4 = n_5 \sin\theta_5$

$$\sin\theta_1 = \frac{n_5}{n_1}\sin\theta_5 = \frac{1.000\ 233}{1.000\ 273}\sin 90°$$

$$\theta_1 = 89.5°$$

(b) No, this is because the image is formed by the reflection of light from the sky at the same fixed angle even if he walks forwards.

▶ Revision 2 Q28 (p.75)

superior mirage 上蜃景

b Fish eye view

Consider a light bulb underwater at point X (Fig 2.2h). Light from the bulb is refracted in all directions in the air through a circular region at the water-air interface. Total internal reflection occurs outside this region. Conversely, light from objects in all directions above the water can reach X, but only through this circular region (Fig 2.2i). Therefore, a diver at X can see objects in all directions above the water, but the view is squeezed into a circle (Fig 2.2j).

This circle is known as Snell's window. ▶

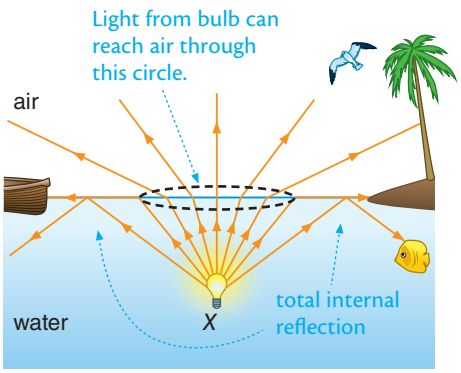


Fig 2.2h *A light bulb underwater.*

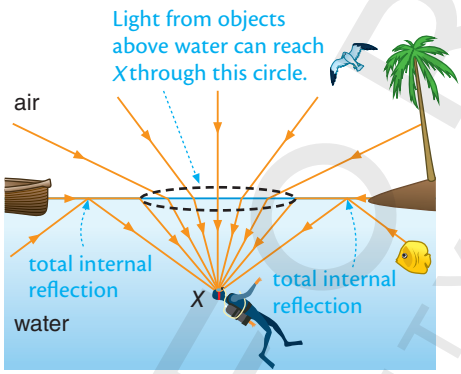


Fig 2.2i *A diver underwater.*



Fig 2.2j *A photograph taken underwater.*

Example 9 | Radius of a diver's view

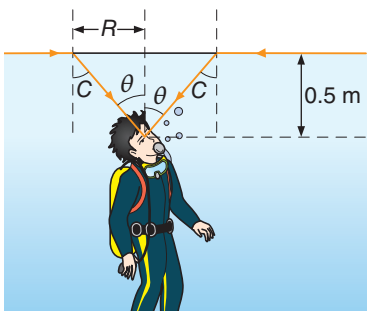
A diver underwater looks upwards. His view is squeezed into a circle. What is the radius R of this circle? His eyes are 0.5 m below the water surface. The refractive index of water is 1.33.

Solution

$$\sin C = \frac{1}{n} = \frac{1}{1.33}$$

$$C = 48.75°$$

$$R = 0.5 \tan 48.75° = 0.570 \text{ m}$$



▶ Revision 2 Q8 (p.69)

Think about...

If the diver is at a deeper position, how will θ and R change?

Know more

View under water

When water is very clean and there is enough light to illuminate objects underwater, images resulting from total internal reflection from the water-air interface together with the circular fish-eye view can be observed.



DSE exam
17(1B)Q7(b), 20(1B)Q7

3 Using total internal reflection

a Optical fibres

An **optical fibre** is as thin as a human hair (Fig 2.2k). It can transmit light from one end to the other (Fig 2.2l).

Video

→ This video shows how a light beam undergoes total internal reflection to go through (a) a simple optical fibre, (b) a bent Perspex rod, and (c) a bundle of optical fibres.

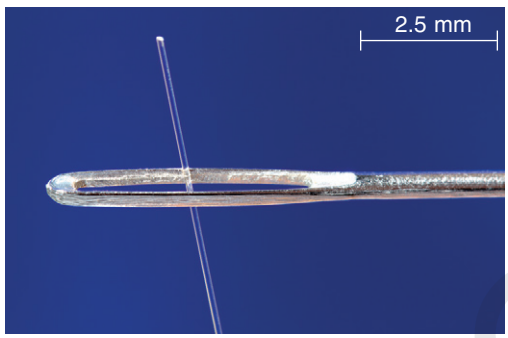


Fig 2.2k An optical fibre passes well through the eye of a needle.

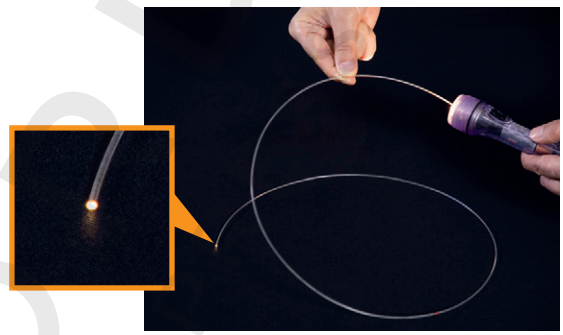


Fig 2.2l Light emerges from the ends of some optical fibres.

Flipped classroom
Watch a video and answer the questions.

It is basically a glass core surrounded by a *cladding* (Fig 2.2m). Since the cladding has a smaller refractive index than the core, total internal reflection can occur when light hits the wall of the core. Therefore, light entering one end of the fibre can be transmitted to the other end.

Total internal reflection can also occur without the cladding, but the cladding protects the core from dirt and damage.

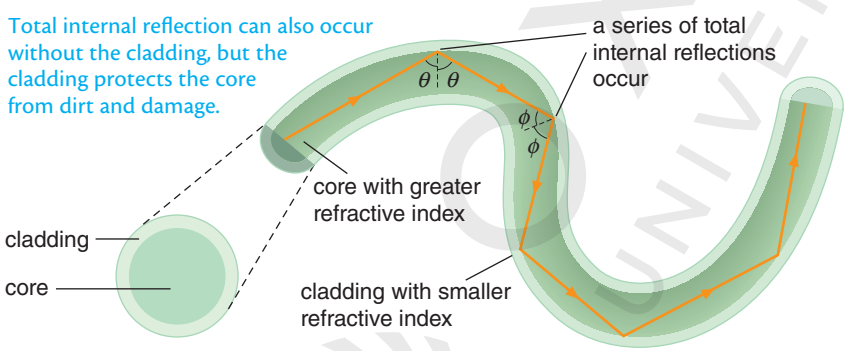


Fig 2.2m Structure of optical fibre.

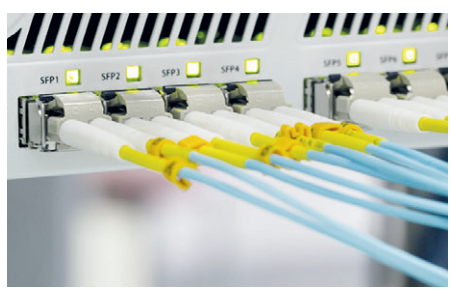


Fig 2.2n Optical fibres for transmitting data.

Cross-link
Endoscopes will be discussed in Book E4 Ch 1. ▶

Optical fibres can transmit data using light signals (Fig 2.2n). They can also be used in endoscopes that allow a doctor to see inside the human body.

Know more

Optical fibres vs copper wires

Optical fibres have the following advantages over copper wires in data transmission:

- Higher data transmission rate
- Lighter, thinner, cheaper and stronger
- Less transmission loss over great distances
- Interference of signals is less likely to occur

The strength of fibreglass is comparable to that of steel. ▶

But the installation and maintenance cost is higher than copper wire.

optical fibre 光纖　　cladding 包覆層

Kao was born in Shanghai. He completed his secondary education in Hong Kong and continued his study and work overseas. He became the Vice-Chancellor (President) of the Chinese University of Hong Kong from 1987 to 1996.

Know more

Father of fibre optics

The idea of using glass fibres for telecommunications was first proposed by *Charles K. Kao* (1933–2018) in 1965. He was awarded the Nobel Prize in Physics in 2009.



OUP eResources

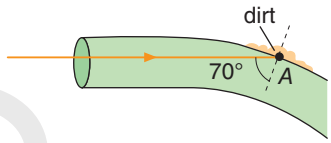
→ This video shows how to make a fluorescent writing board.

You can make your own fluorescent writing board easily by watching the following video.

Video

Example 10 | Use of cladding in optical fibres

A simple optical fibre is made of a plastic core with no outer cladding. The refractive index of the plastic is 1.62. A light ray is incident normally from one end of the fibre and reaches point A as shown.



(a) A dirt patch of refractive index 1.7 is accidentally attached to the outside of the optical fibre at point A. Explain whether total internal reflection will occur or not when the ray reaches point A.

(b) If the core is coated with a cladding of refractive index 1.52, will total internal reflection occur when the ray reaches point A?

Solution

(a) Total internal reflection will not occur. The dirt has a greater refractive index than the core, so light is refracted out to the dirt.

(b) By Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$1.62 \times \sin C = 1.52 \times \sin 90°$$

$$C = \sin^{-1}\left(\frac{1.52}{1.62}\right) = 69.8°$$

Since the critical angle is smaller than 70°, total internal reflection will occur when the ray reaches point A.

Exercise 2.2 Q4 (p.66)

Know more

Fluorescent writing board

Have you ever seen a fluorescent writing board (Fig a)? Words written with fluorescent ink glow brightly. Do you know how it works?

The writing board consists of a glass plate and light sources on its sides. The light bounces back and forth inside the glass plate by total internal reflection (Fig b). Since the critical angle of the glass-ink interface is much larger than that of the glass-air interface, light reaching the fluorescent ink leaks from the glass more easily. This makes the words written with the ink glow.

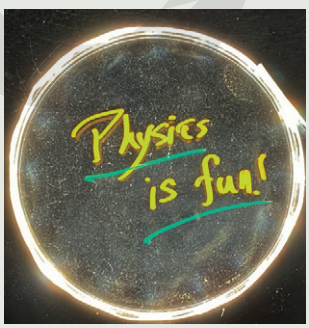


Fig a

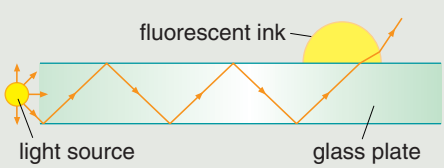
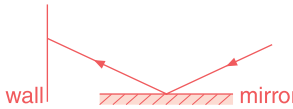


Fig b

Charles K. Kao 高錕

Ts may demonstrate the formation of multiple images by simply directing a laser beam to a mirror at an angle. Multiple light spots will be formed on the wall.



→ This simulation shows that in use of total internal reflection, how a ray goes through a maze with the help of prisms. The simulation is available on OUP web in PC version (.exe) only.

→ This video shows how a light ray travels in a prism.

Simulation
Video

b Using prisms as mirrors

When light from an object falls on a glass mirror, multiple reflections occur between the top and the bottom surfaces of the glass (Fig 2.2o). Multiple images are formed and make the main image I_2 less clear.

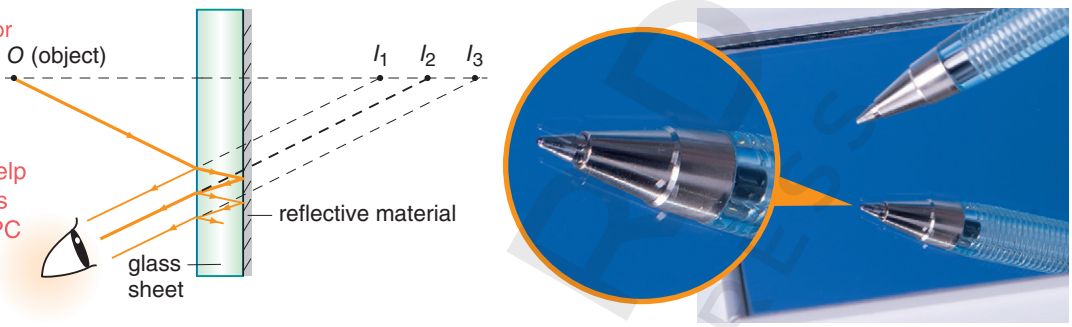
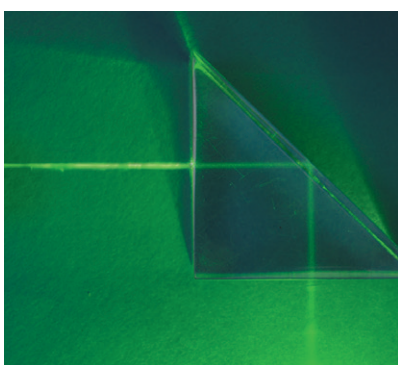


Fig 2.2o *Multiple images formed by a glass mirror.*

The problem can be solved by replacing the glass mirror with a prism. Light is totally internally reflected by the inside surface of a prism if the angle of incidence is greater than the critical angle (Fig 2.2p). In this way, a prism behaves like a perfect mirror (Fig 2.2q).

The intensity of the reflected light is stronger by using prisms.



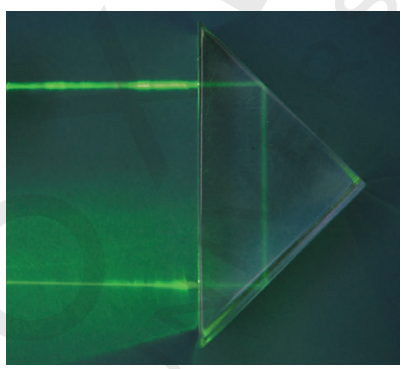


Fig 2.2p *Total internal reflection in 45°-90°-45° prism.*



Fig 2.2q *A prism as a perfect mirror.*

→ This is a Video Quiz about the experiment 'Reflection in a prism'.

Many optical instruments, e.g. binoculars and periscopes, use prisms instead of plane mirrors to produce quality images (Fig 2.2r).

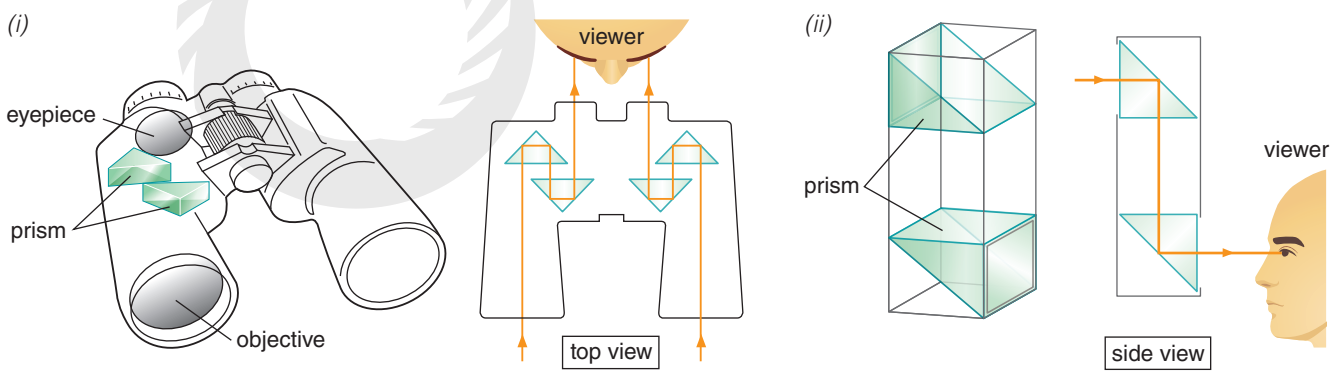


Fig 2.2r *Prism arrangement in (i) binoculars and (ii) periscope.*

Headlight reflectors in cars and rear reflectors in bicycles also use prisms (Fig 2.2s). The reflector reflects the light emitted by the car behind backwards so that the driver can see the vehicle in front at night (Fig 2.2t).

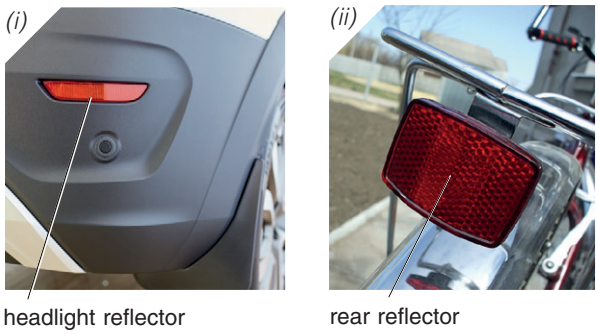


Fig 2.2s *(i) A headlight reflector on a car and (ii) a rear reflector on a bicycle.*

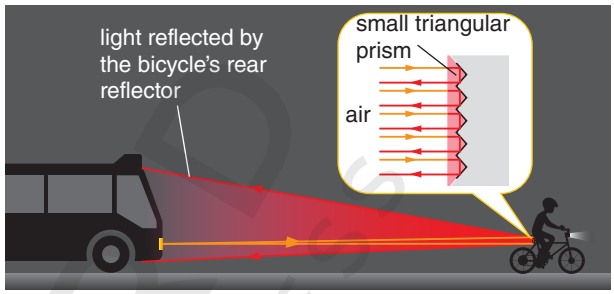


Fig 2.2t *Light is reflected by the reflector.*

C Sparkling of diamonds

A diamond has a large refractive index and the critical angle for a diamond-air interface is small. When it is cut into a suitable shape, it can reflect most of the incident light through its top surface by total internal reflection (Fig 2.2u). This makes a diamond sparkle (Fig 2.2v).

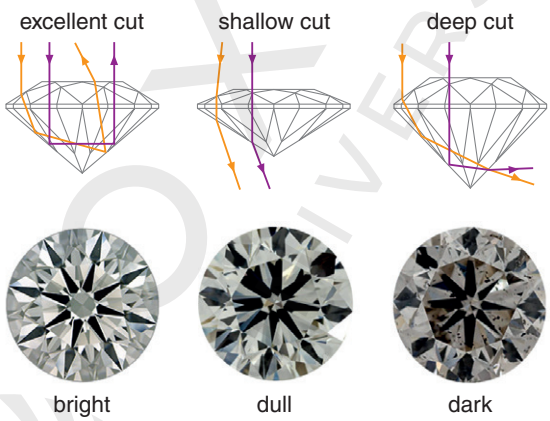


Fig 2.2u *What the diamond will look like with different cuts.*



Fig 2.2v *A diamond sparkles after it is cut properly.*

For diamond,

$\sin C = \frac{1}{n} = \frac{1}{2.42}$

$C = 24.4°$

For glass,

$\sin C = \frac{1}{n} = \frac{1}{1.5}$

$C = 41.8°$

▶ If a piece of glass is cut in the same way as a diamond, light will leak through its bottom as the critical angle of a glass-air interface is much larger (Fig 2.2w). Therefore, it does not sparkle as much as a diamond.

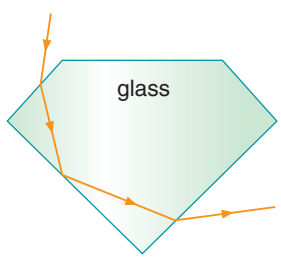


Fig 2.2w *Light leaks through a piece of glass.*

Exercise 2.2

★**1** Which of the following statements are correct when total internal reflection occurs?

(1) The angle of incidence must equal the critical angle.

(2) The light meets an optically less dense medium.

(3) The angles that the incident and reflected rays make with the normal are the same.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**2** A ray of light enters and leaves a prism normally as shown (Fig a). Which of the following statements must be correct?

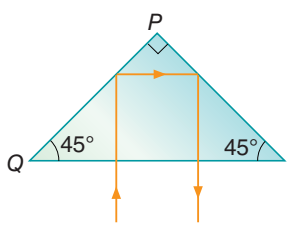


Fig a

(1) The refractive index of the prism is 1.41.

(2) The critical angle of the prism is 45°.

(3) If the prism is replaced by another one with a greater refractive index, total internal reflection will still occur at face *PQ*.

A (1) only B (3) only

C (1) and (2) only D (1), (2) and (3)

★**3**

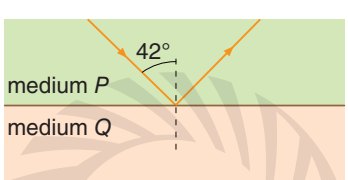


Fig b

Total internal reflection occurs when a ray of light travels from medium *P* to medium *Q* (Fig b). Which of the following statements must be correct?

(1) *P* has a greater refractive index than *Q*.

(2) The critical angle of this interface is larger than 42°.

(3) Total internal reflection will occur if a ray of light travels from *Q* to *P* with an angle of incidence of 42°.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**4** An optical fibre has a glass core without any cladding. A ray of light is totally internally reflected down the fibre, striking the walls at angle θ (Fig c). The refractive index of the glass core is 1.65.

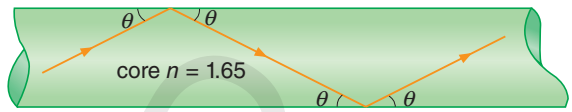


Fig c

(a) Find the critical angle of the glass-air interface.

(b) What is the maximum value of θ for which the ray will be retained within the fibre?

(c) How will the critical angle change if the glass core is surrounded by a cladding of refractive index 1.4? No calculation is needed.

(d) Why should the glass core be surrounded by a cladding?

★**5** The critical angle for the interface between medium *A* and medium *B* is 44.9°. The refractive index of medium *A* is 1.7. Find the refractive index of medium *B*. Assume medium *A* has a greater refractive index than medium *B*.

★**6** A ray of light enters a rectangular glass block from side *AB* (Fig d). When $i = 60°$, the ray is totally internally reflected from side *BC* and then emerges from side *CD*.

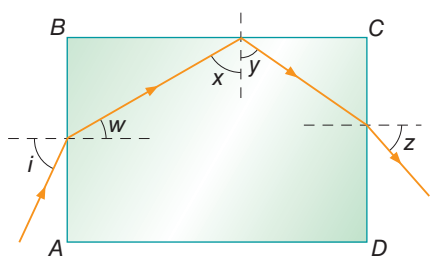


Fig d

The refractive index of the glass block is 1.5.

(a) Calculate the angles *w*, *x*, *y* and *z*.

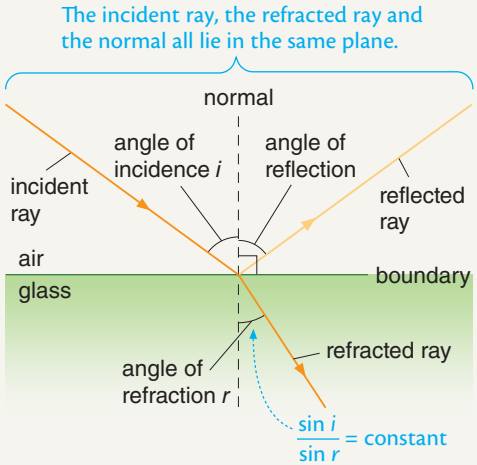
(b) Compare the angle of emergence *z* of the ray with the angle of incidence *i*.

(c) Will the ray emerge from side *BC* if *i* decreases gradually? Explain briefly.

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Key concepts 2

1 The laws of refraction

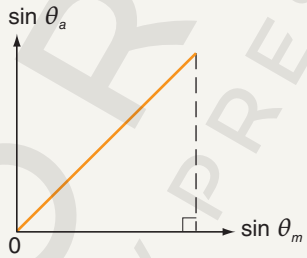


2 Refraction index (n)

it is a ratio ⇒ no unit

$$n = \frac{\text{speed of light in air}}{\text{speed of light in a medium}}$$

or

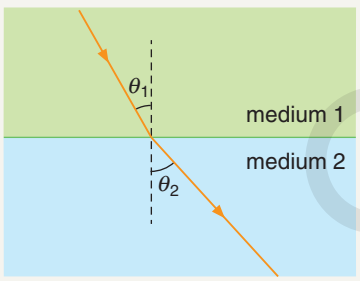


n = slope = $\frac{\sin \theta_a}{\sin \theta_m}$

θ_a: angle between ray and normal in air

θ_m: angle between ray and normal in medium

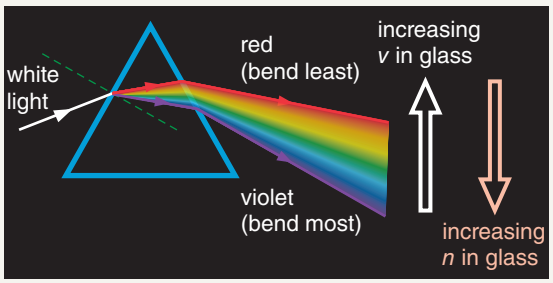
3 General form of Snell's law



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

4 Dispersion

Different coloured lights travel at the same speed in air but slightly different speeds in other media.

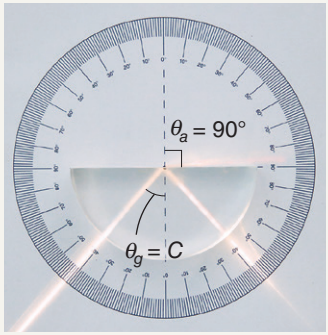


5 Total internal reflection

1 Total internal reflection occurs when
- light travels from an optically denser medium to an optically less dense medium, and
- the angle of incidence is larger than the critical angle C.

2 For a medium of refractive index n, the critical angle of the medium-air interface is given by

$$\sin C = \frac{1}{n}$$



Key terms

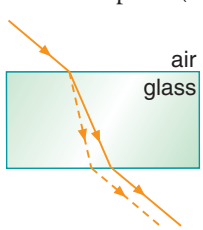
1 angle of refraction 折射角 p.36
2 critical angle 臨界角 p.55
3 dispersion 色散 p.51
4 laws of refraction 折射定律 p.42
5 medium 介質 p.36
6 optical fibre 光纖 p.62
7 refracted ray 折射線 p.36
8 refraction 折射 p.36
9 refractive index 折射率 p.43
10 Snell's law 斯涅耳定律 p.42
11 total internal refraction 全內反射 p.54

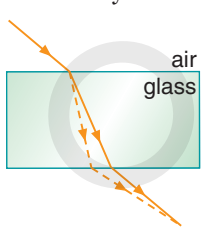
Revision 2

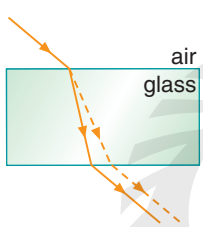
Multiple-choice questions

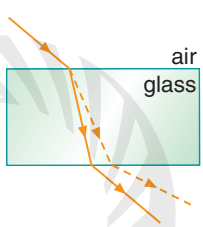
Unless otherwise specified, take the speed of light in air as 3×10^8 m s^{-1}.

★1 2.1 A ray of light in air travels through a rectangular glass block. If the glass is replaced by one of higher refractive index, which of the following diagrams best show the original path (dotted line) and the new path (solid line) of the ray?

A


B


C


D


★2 2.1 A ray of light travels from oil to glass (Fig a). The refractive index of glass is n_g. The speed of light in vacuum is c.

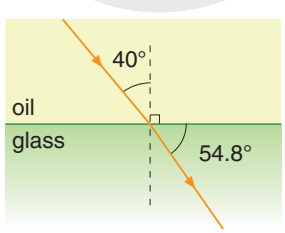


Fig a

What is the speed of light in oil?

A $\frac{0.90c}{n_g}$
B $\frac{1.12c}{n_g}$
C $\frac{1.27c}{n_g}$
D $\frac{1.33c}{n_g}$

★3 Misc A girl is looking at a shark in aquarium (Fig b). Which of the following statements is/are correct?



Fig b

(1) To the girl, the shark appears nearer than it really is.
(2) To the shark, the girl appears further away than she really is.
(3) The shark is unable to see the girl as total internal reflection occurs.

A (1) only
B (2) only
C (1) and (2) only
D (1) and (3) only

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(For Q4–5.) A ray of light travels from air to media P, Q and R in turn. Different sets of angles of incidence i and the corresponding angles of refraction r are measured. Figure c shows the graphs of sin i against sin r for the three media.

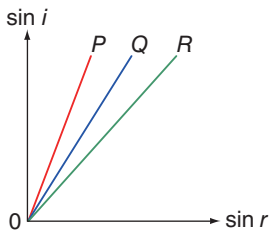


Fig c

★4 2.1 Which of the following statements is/are correct?

(1) P has a greater refractive index than Q.

(2) R is optically denser than Q.

(3) Light travels slower in R than in P.

A (1) only B (3) only

C (1) and (2) only D (1), (2) and (3)

★5 2.2 Total internal reflection would **never** occur when light travels from

A P to Q. B P to R.

C Q to R. D R to Q.

★6 2.2 An observer sees the image of a letter F through a right-angle prism (Fig d).

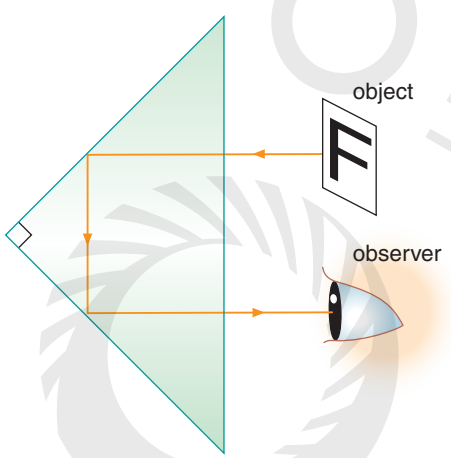


Fig d

Which image will the observer see?

A  B 

C  D 

★7 2.2 A ray of light travels as shown (Fig e).

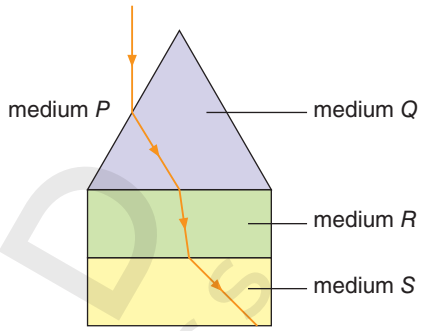


Fig e

In which of the following cases may total internal reflection occur?

(1) Light travels from R to P.

(2) Light travels from Q to S.

(3) Light travels from S to R.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★8 2.2 A diver is looking at the air-water interface, of which the critical angle is C (Fig f).

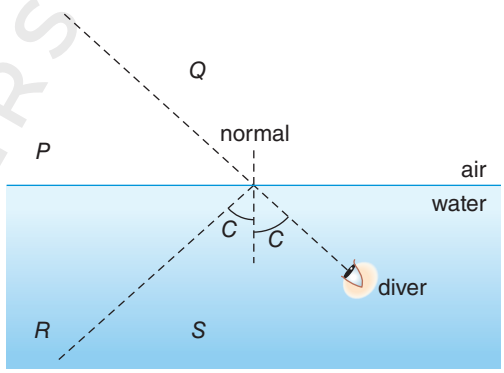


Fig f

The diver cannot see objects in region

A P. B Q.

C R. D S.

★★9 Misc Figure g shows a light ray entering a spherical water droplet from air at point P and reaches point Q. O is the centre of the droplet. The refractive index of water is n.

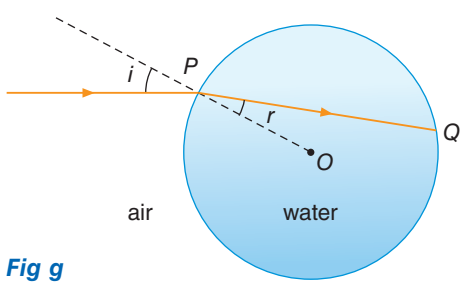


Fig g

Which of the following are correct?

(1) $\sin i = n \sin r$

(2) Total internal reflection will never occur at point P.

(3) Total internal reflection will never occur at point Q.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

Refer p.135

**10 ★★ 2.2 Figure h shows a straight optical fibre. The reflective indices of the core and the cladding are n_1 and n_2 respectively, and $n_1 > n_2$.

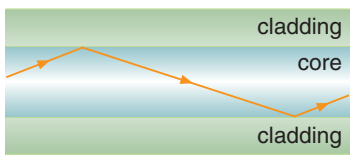


Fig h

For a ray of light passing from one end of the fibre to the other end without leaking from the core, if the minimum time needed is T, what is the maximum time needed?

A $\frac{T}{n_1}$ B $\frac{T}{n_2}$

C $\frac{n_1}{n_2}T$ D $\frac{n_2}{n_1}T$

Refer p.62, 63

**11 ★★ 2.1

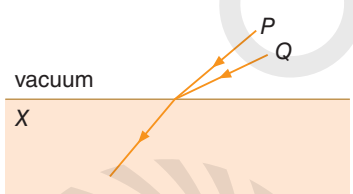


Fig i

Two coloured lights P and Q strike medium X at the same point from a vacuum and then travel in medium X along the same path as shown (Fig i). Which of the following statements is/are correct?

(1) Dispersion occurs.

(2) P and Q travel with the same speed in X.

(3) The reflective index for P in X is smaller than that for Q.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

Refer p.51, 52

Q12 Exam Report: More than one-third of the candidates thought that θ depends on the refractive index of glass (option D).

**12 HKDSE 2016 Paper 1A Q17 2.2

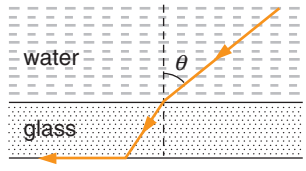


Fig j

A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle θ on the glass sheet and finally emerges into air along the glass-air interface as shown (Fig j). Find θ.

Given: refractive index of water is 1.33.

A 41.2°

B 48.8° (48%)

C 53.1°

D It depends on the refractive index of glass.

**13 HKDSE 2016 Paper 1A Q20 2.1

A beam of white light is separated into different colours after entering a glass prism because lights of different colours

A are diffracted to different extents by the prism.

B undergo total internal reflection at different angles inside the prism.

C travel at different speeds in vacuum.

D travel at different speeds in glass. (59%)

Conventional questions

*14 ★ 2.1 Figure k shows a ray of violet light travelling from air to a plastic prism of refractive index 1.4.

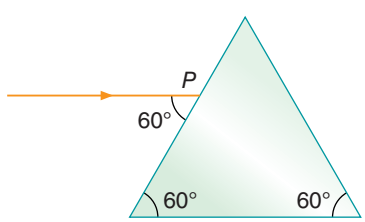


Fig k

(a) Find the angle of refraction at P. (1 mark)

(b) Sketch the paths of the rays in Figure k. (2 marks)

(c) If the ray of violet light is replaced by a ray of white light, what can be observed when the ray emerges from the prism? (1 mark)

★15 Misc Figure l shows a ray of light entering water from air. The refractive index of water is 1.33.

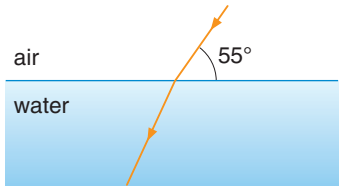


Fig l

(a) Find the angle of refraction. (2 marks)

(b) Find the critical angle for the air-water interface. (2 marks)

(c) Will total internal reflection occur if the angle of incidence is larger than the one obtained in (b)? Explain briefly. (2 marks)

★16 Misc In Figure m, the part of the Pyrex rod in the oil seems to disappear. The refractive indices of water and oil are 1.33 and 1.47 respectively.

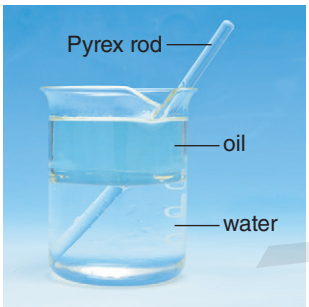


Fig m

(a) Estimate the refractive index of Pyrex. (1 mark)

(b) Estimate the speed of light in Pyrex. (2 marks)

(c) State the conditions for total internal reflection to occur at the oil-water interface. (3 marks)

★17 Misc (a) A common plane mirror consists of a glass sheet coated with silver on the back surface. When a light beam from an object falls on the mirror, it travels as shown in Figure n and several images of the object are formed.

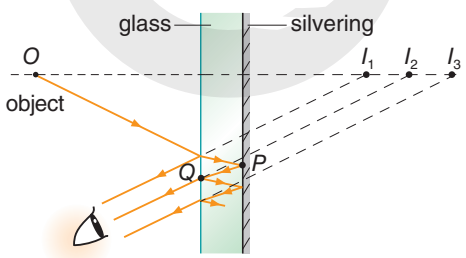


Fig n

According to Figure n, name the phenomena that occur at P and Q. (2 marks)

(b) A prism can be used as a 'mirror'.

(i) Which phenomenon is involved in using a prism as a 'mirror'? (1 mark)

(ii) Draw a diagram to show how a prism can be used as a 'mirror'. (2 marks)

(iii) Suggest an advantage of a prism over a mirror in forming images. (1 mark)

★18 2.1 In an experiment on the laws of refraction, a ray of light passes from air into a block of Perspex. The results are listed in Table a.

Angle of incidence i	Angle of refraction r
10°	8°
20°	15°
30°	20°
40°	26°
50°	31°
60°	36°
70°	39°
80°	42°

Table a

(a) Plot a suitable graph to find the refractive index of Perspex. (4 marks)

(b) It is not preferred to use a particular pair of data to calculate the refractive index of Perspex. Explain why. (2 marks)

★19 Misc A cylindrical container is mounted on a vertical wall. O is the centre of the container. A ray of light is directed at the liquid as shown (Fig o). The angle θ is changed and the corresponding values of angle ϕ are recorded. Figure p shows how sin ϕ varies with sin θ.

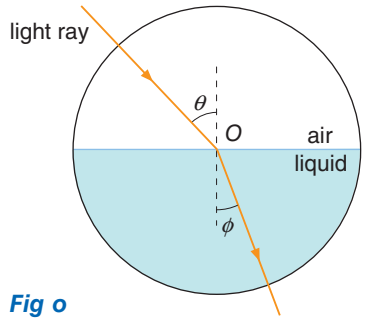


Fig o

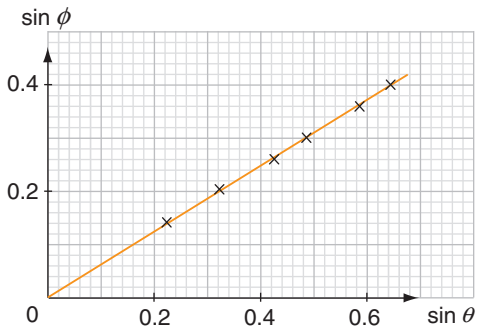


Fig p

(a) Find the refractive index of the liquid. (2 marks)

(b) Find the critical angle for the interface between the liquid and the air. (2 marks)

(c) Another ray of light is directed vertically upwards to the liquid as shown in Figure q. Assume that the container wall and the liquid have similar refractive indices.

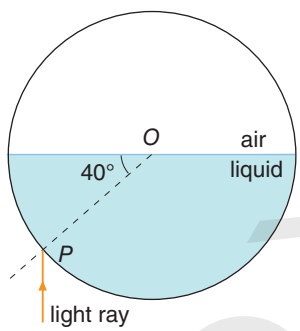


Fig q

(i) Will total internal reflection occur at P? Explain briefly. (2 marks)

(ii) Sketch the path of the light ray just after hitting P in Figure q. (1 mark)

★20 Misc A light beam is directed at a tank at an angle. The tank contains some sugar solution with a higher concentration near the bottom. The light beam bends downwards when passing through the solution (Fig r).

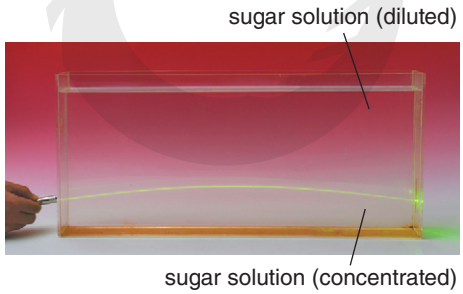


Fig r

(a) Describe how the refractive index of the sugar solution changes with the concentration. (1 mark)

(b) The light beam now moves to a higher position and points down at an angle (Fig s).

(i) Explain whether total internal reflection will take place in the sugar solution. (3 marks)

(ii) How will the beam bend? Sketch the path in Figure s. (1 mark)



Fig s

★21 Misc In Figure t, a ray of light enters an unclean diamond. The ray is reflected internally and finally hits a facet at an angle of incidence of 25°. A tiny oil droplet adhered to the diamond at point O. Take the critical angle for a diamond-air interface to be 24.4° and the refractive index of the oil to be 1.40.

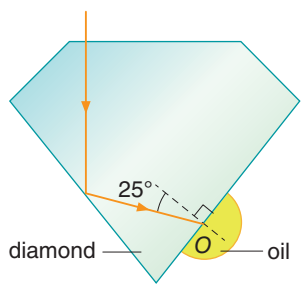


Fig t

(a) Find the refractive index of the diamond. (2 marks)

(b) Find the critical angle for the diamond-oil interface. (2 marks)

(c) Briefly explain why the diamond loses its brilliance when the oil droplet adheres to it. (2 marks)

(d) Complete the path of the ray in Figure t. Assume that the oil droplet is hemispherical with centre O. (2 marks)

★22 2.1 Danny carries out an experiment to verify Snell's law.

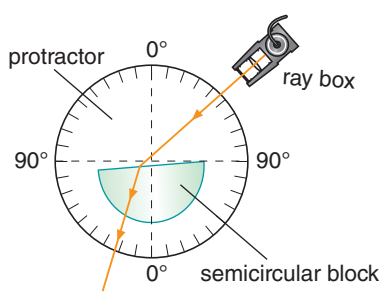


Fig u

(a) State **two** mistakes made by Danny during the experiment as shown in Figure u. (2 marks)

(b) State another error, as shown in Figure v, which is caused by the apparatus. (1 mark)

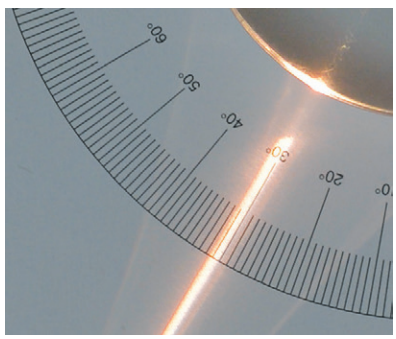


Fig v

(c) The teacher asks Danny to direct the ray from both the left and right sides of the block for each angle of incidence (Fig w). State the advantage of doing so. (2 marks)

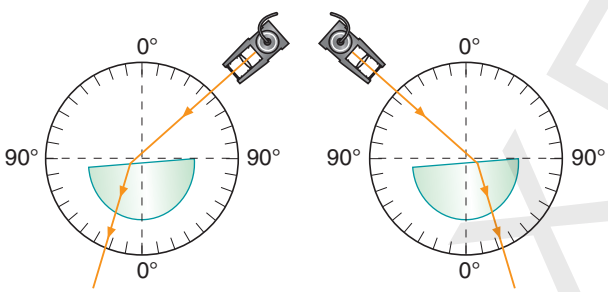


Fig w

★**23** Misc Fig x shows a gadget called 'lazy glasses'. A person wearing the glasses can read a book for a long time without leaning the head down and causing neck strain.



Fig x

(a) Cindy would like to make a similar gadget by using a 45°-90°-45° glass prism. She puts the prism above a paper with a letter 'F' (Fig y). A light ray from the letter 'F' incidents normally on one side of the prism. The refractive index of the glass is 1.64.

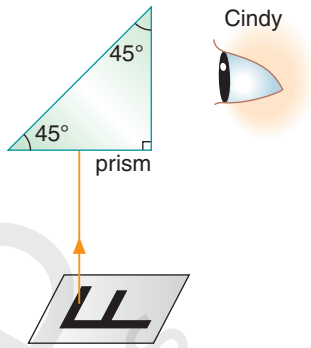


Fig y

(i) Find the critical angle of the glass-air interface. (2 marks)

(ii) Complete the path of the light ray until it finally emerges from the glass prism. (2 marks)

(iii) Show that this design is not appropriate by sketching the shape of the image seen by Cindy. (1 mark)

(b) To improve the glasses, Cindy modifies the design by using a 30°-90°-60° glass prism and adding a mirror facing down on top of the prism (Fig z). A light ray from the letter 'F' incidents normally on one side of the prism.

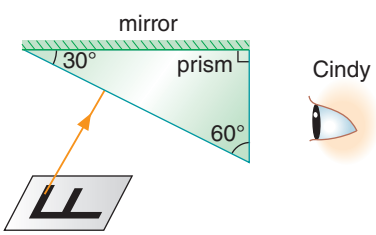


Fig z

(i) Complete the path of the light ray until it finally emerges from the glass prism. (2 marks)

(ii) Show that the new design solves the problem stated in (a)(iii) by sketching the shape of the image seen by Cindy. (1 mark)

★**24** Misc You are provided with the following apparatus (Fig aa).

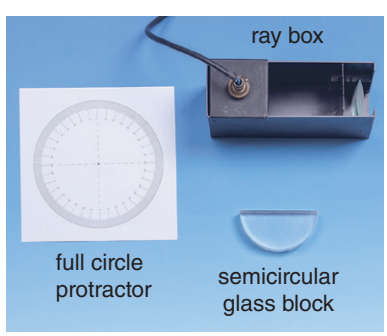


Fig aa

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(a) Describe how you should use the apparatus to conduct an experiment to verify the relation between i and r as described by Snell's law. (5 marks)

(b) Using the same set of apparatus, describe how you can find the critical angle of the glass-air interface. (3 marks)

**25 2.1 To add privacy while still allowing light to pass through, ground glass (Fig ab) or gauze curtain (Fig ac) is often used in windows.

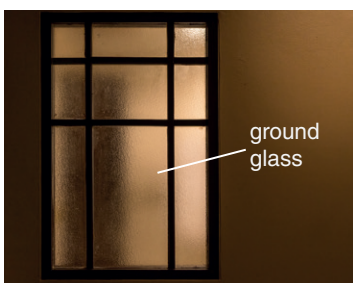


Fig ab



Fig ac

Parallel rays of light fall on a ground glass window pane as shown in Figure ad.

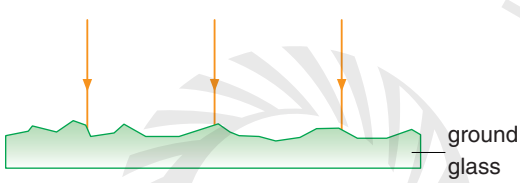


Fig ad

(a) Sketch the paths of the rays in Figure ad. (1 mark)

(b) Explain the working principle of ground glass windows. (3 marks)

(c) State an advantage of using ground glass over gauze curtain in windows. Explain why. (2 marks)

Refer p.36

**26 Misc A spherical air bubble is trapped inside the glass layer of a drinking glass (Fig ae). The reflective index of the glass is 1.62.

A ray of light enters the bubble as shown (Fig af). Point O is the centre of the air bubble.



Fig ae

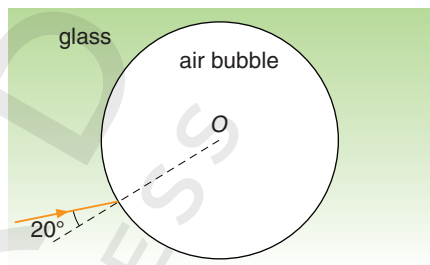


Fig af

(a) Find the angle of refraction inside the air bubble. (2 marks)

(b) Find the maximum angle of incidence at which the ray can enter the air bubble. (2 marks)

(c) Explain whether the ray will undergo total internal reflection inside the air bubble. (3 marks)

(d) Complete the path of the ray in Figure af. (2 marks)

Refer p.57, 135

27 Misc SQA Higher 2016 Q10

Retroflective materials reflect light to enhance the visibility of clothing (Fig ag).



Fig ag

One type of retroflective material is made from small glass spheres partially embedded in a silver-coloured surface that reflects light.

A ray of monochromatic light follows the path shown as it enters one of the glass spheres (Fig ah).

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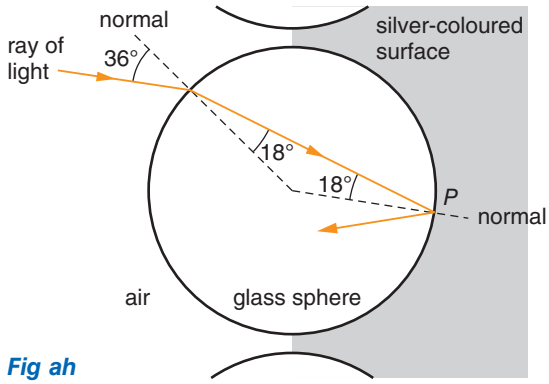


Fig ah

(a) Calculate the refractive index of the glass for this light. (3 marks)

(b) Calculate the critical angle for this light in the glass. (3 marks)

(c) The light is reflected at point *P*. Complete the diagram to show the path of the ray as it passes through the sphere and emerges into the air. (1 mark)

Exam report: In question (c) most candidates could not identify that the ray of light would emerge parallel to the original ray of light.

See p.76 for the Exam Report.

28 HKDSE 2015 Paper 1B Q6

2.2

Read the following description about a **mirage** and answer the questions that follow.

A mirage is often seen on highways during hot summers. Pools of water seem to cover the roadway far ahead. Distant objects appear to be reflected by the surface of the 'water'. The phenomenon is caused by the difference in refractive index between the hot air near the road surface and the cooler air above it. The refractive index of cool air is greater than that of hot air, but the differences are so small that the subsequent deviations of light rays are tiny. Sufficiently large temperature differences between the hot air near the road surface and the above cooler air over a short height (i.e. high temperature gradient) and light rays travelling along sufficiently long path lengths are required to form a mirage.



Fig ai Mirage seen on a highway. This photo was taken with a telephoto lens which gives the perception that the viewer is very close to the car ahead.

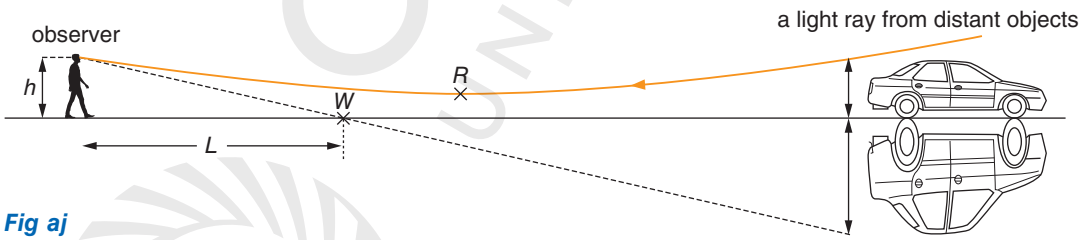


Fig aj

Figures aj and ak illustrate the principle of the phenomenon. Air of different temperatures is simplified to several layers and modelled as parallel slabs as shown in Figure ak. The bending of the light ray from distant objects is much exaggerated. θ_1, θ_2, θ_3, and θ_4 denote the angles of incidence at various boundaries of air layers.

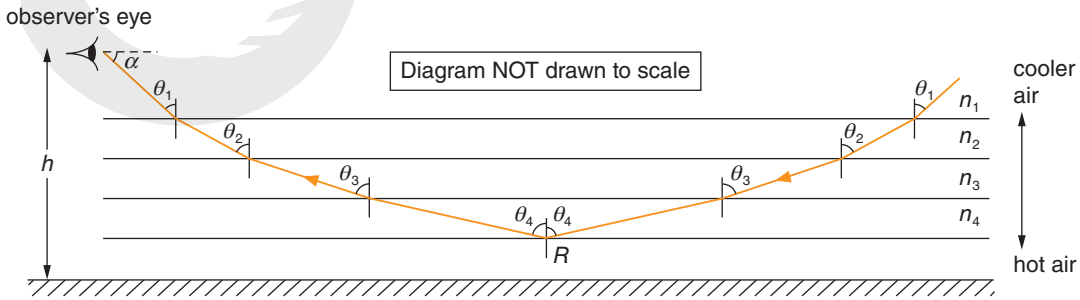


Fig ak

(a) State **ONE** essential condition for a mirage to be observed. (1 mark)

(b) (i) Referring to Figure ak, deduce the relationship between θ_1, θ_4, and refractive indices n_1, n_4. For total internal reflection just to occur at R, θ_4 can be taken as 90°. Hence, find the corresponding value of θ_1 if $n_1 = 1.000261$ and $n_4 = 1.000221$. (3 marks)

(ii) Find L in Figure aj if $h = 1.5$ m. (Note: $\alpha + \theta_1 = 90°$ in Figure ak.) (2 marks)

(c) A thirsty traveller in a vast desert sees similar mirages such that a 'water source' appears at W which is distance L away like the one in Figure aj. If he walks a distance L towards that 'water source', how far would the 'water source' appear to him? Explain your answer. (2 marks)

29 HKDSE 2017 Paper 1B Q7

Misc

(a) A light ray enters a rectangular plastic block $ABCD$ from the air at point E, and the angle of incidence is θ. The light ray emerges along face BC as shown in Figure al. The refractive index of the plastic is 1.36.

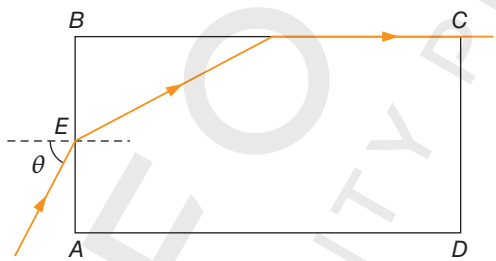


Fig al

(i) Find the critical angle of the plastic. (2 marks)

(ii) Find the value of θ. (3 marks)

(iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ, sketch the path of light ray in Figure al. (2 marks)

(b) A student designs a periscope using two plastic prisms, the refractive index of plastic is 1.36. As shown in Figure am, an object is placed in front of the periscope.

Q28 Exam report: This question was based on a passage describing the formation of a mirage. The situation was unfamiliar to candidates and their general performance was unsatisfactory. In (a), few were able to state correctly an essential condition for a mirage to be observed based on the information given. Parts (b)(i)(ii) were well answered. A few candidates did not know how to find the value of θ_1 in (b)(i) using the relation $n \sin \theta$ = constant. Candidates answered part (c) poorly. Only the most able ones explained correctly why the 'water source' remained a distance L away.

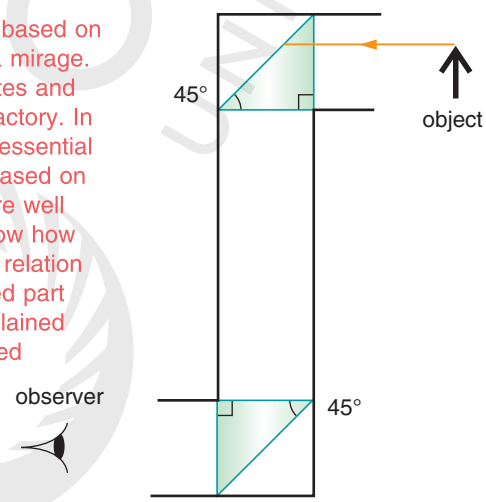


Fig am

Q29 Exam report: This question was well answered. Most were able to obtain the correct answers in parts (a)(i)(ii). Some candidates failed to recognize that in (a)(iii) the light ray would eventually emerge from the plastic block as the incident angle within the block becomes smaller than the critical angle. Weaker ones drew a light ray bending towards the normal when the light ray emerged from the block to the air. Part (b) was in general well answered.

(i) Complete the path of the light ray from the object in Figure am, and explain why the periscope fails to work. (3 marks)

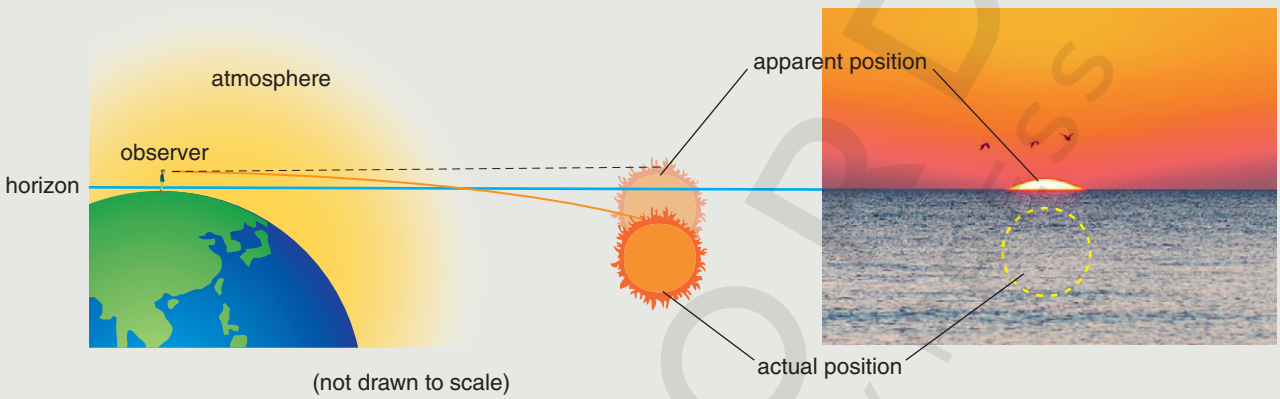
(ii) What can be used to replace the two plastic prisms so that the periscope can work properly? (1 mark)

Know more

Refraction in atmosphere (p.39)

Sunrise

Do you know that you can see the sun a few minutes before it reaches the horizon? Air is less dense at higher altitude, and light travels slightly faster in less dense air. During sunrise, sunlight is continuously refracted by the atmosphere, so the sun appears higher.



Distorted images formed by hot air

Light travels slightly faster in hot air than in cold air. When light rays from an object pass through unstable hot air, they are refracted irregularly before entering our eyes. As a result, the object appears distorted.

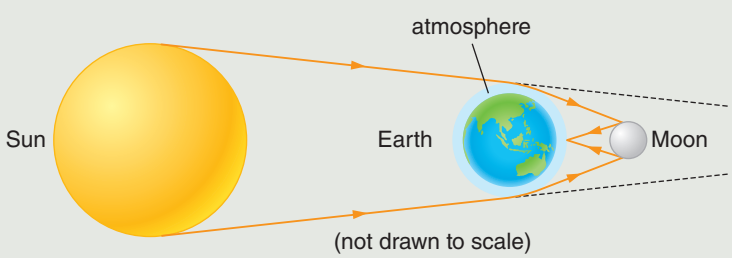




Lunar eclipse

When the Moon is completely inside the *umbra* of the Earth, total lunar eclipse occurs. The Moon is still visible because sunlight is refracted by the Earth's atmosphere to the Moon.



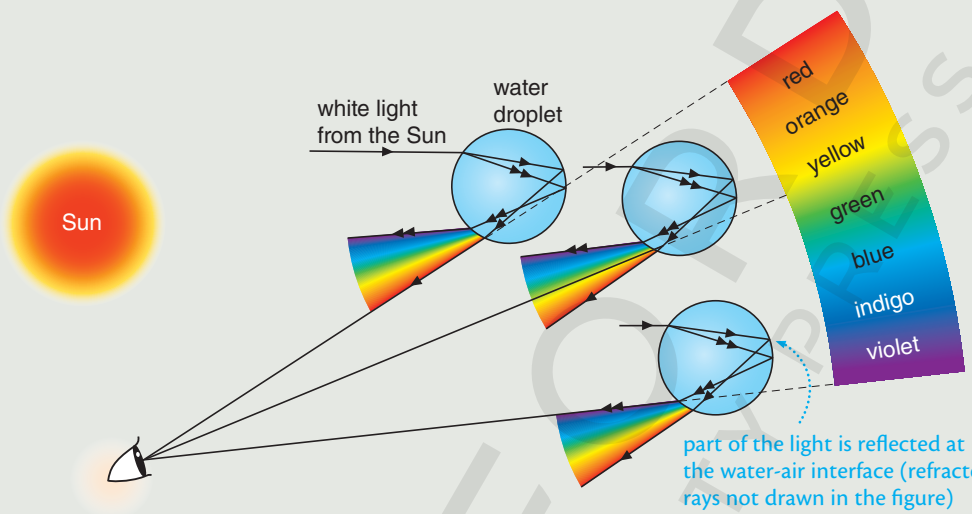


umbra 本影區

Know more

Rainbow (p.51)

A rainbow may be formed after rain. This is due to the dispersion of sunlight. Just after it has been raining, there are many water droplets in the air. When the sun shines on the water droplets in the air, dispersion occurs due to refraction and reflection. Each droplet produces a complete colour spectrum, but only one colour from each droplet reaches our eye. As a result, we see a rainbow with red on the top and violet at the bottom.



Disappearing magician (p.54)

The magic can be explained by total internal reflection.

① A layer of air is trapped between the two plastic cups.

② When light rays from the magician on the inner cup travels from air to water, they bend towards the normal.

③ Most refracted rays reach the top water-air interface at an angle larger than the critical angle. These rays are totally reflected downward and do not reach an observer above the water.

④ The drawing can be observed if an observer looks at the cup at a position below the water surface.

